

TOWARDS A CALCULUS FOR NON-LINEAR SPECTRAL GAPS

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ABSTRACT. A new proof of a result of Lafforgue on the existence of expanders on uniform convex Banach spaces is given. Along the way, we develop abstract analysis of the zigzag product, the spectral decay of powers and generalize the Khot-Naor the quotient of the Hamming cube construction into quotient of “noisy cube” that does not embed in K -convex spaces.

1. INTRODUCTION

Expanders were first defined by Bassalygo, Pinsker and Margulis [21, 7, 17] inspired by the work of Gallager on parity-check codes [10]. They were introduced to Computer Science in the work of Valiant on super concentrators [31]. Since then they have been used extensively. Much of the progress was in building expanders with better parameters and more explicit constructions.

Zigzag expanders were developed in [26]. It has since found many applications [9, 24, 27, 1]. For general introduction to expander graphs, see [29]. MM
 «To add a bit of the history of expanders, previous constructions, random graphs. —Manor»

Expanders were first used as examples for badly embeddable metric spaces in [16], and then expanded upon in [18].

The authors showed a dichotomy result [19] about the asymptotic of distortions attainable when embedding all metric spaces. Motivation for this problem comes from the paper [4]. The best dichotomy currently known is either distortion is 1, or it is at least $\log^\beta n$ for some $\beta > 0$ that depends on the host space. However of all spaces in which the distortion is not 1, the best distortion we have is $O(\log n)$. The work in this paper may be a step forward in improving the dichotomy result.

Lafforgue proved the existence of expanders w.r.t. uniform convex (and even K -convex) spaces before [14]. He was advancing a program of Gromov [11] that showed counter-examples to the approach of Yu [32], to a counter example for Kasparov and Yu [12].

Our approach for base graphs begins with Khot and Naor [13] construction of non-embeddable metric in L_1 . The analysis uses ideas of Pisier [23].

The zig-zag construction requires using power to increase the spectral gap. We could generalize it to uniformly convex spaces, and $CAT(0)$ spaces, but only partially, replacing the power with the weaker (but still sufficient) operation of Sarkozy averages. Our approach is inspired by the Markov cotype of Ball [5].

Markov type and Markov cotype have been the focus of many recent studies on the geometry of metric spaces, e.g. [6, 20, 8]. We generalize Ball’s result that p -convex spaces has Markov cotype p to certain notion of barycentric spaces that includes $CAT(0)$ spaces. We do this by extending Pisier’s Martingale cotype inequality for this class of metric spaces. The use of martingales in $CAT(0)$ spaces was pioneered in [30]. A side effect of our result that $CAT(0)$ has Markov cotype 2 is that Ball’s Lipschitz Extension theorem now works for $CAT(0)$ spaces as targets.

2. PRELIMINARIES

2.1. Definitions. All graphs in this paper, unless explicitly written otherwise, are undirected, connected, regular, multi-graph (meaning that we allow multiple edges and self loops). A graph without multiple edges and self loops is referred to as simple graph. A graph $G = (V, E)$, where V is the set of vertices, and E is the *multiset* of *directed* edges. Since all our graphs are undirected each edge between $x \in V$ and $y \in V$ appears

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as two ordered pairs $(x, y), (y, x) \in E$. A self-loop will appears *twice* in E . As E is multiset, the union of two multisets takes into account the multiplicity.

An (n, d) graph is a connected d -regular graph on n vertices. Let $1 > \lambda_2 \geq \dots \geq \lambda_n \geq -1$ be the eigenvalues of the random walk matrix associated with G . As customary in spectral graph theory, we write $\lambda = \max\{\lambda_2, -\lambda_n\} = \max_{i \geq 2} |\lambda_i|$.

Let $\tau = (\{0, 1\}, \{(0, 1), (1, 0)\})$ be the graph consisting of a single edge. For an undirected (multi)graph $G = (V, E)$ the double cover of G is defined to be the tensor graph product

$$G \otimes \tau = (V \times \{0, 1\}, \{((u, 0), (v, 1)), ((v, 1), (u, 0)) : (u, v) \in E\}).$$

Definition 1 (Kernel & Metric). A symmetric kernel ρ on X is a function $\rho : X \times X \rightarrow [0, \infty)$ satisfying for every $x, y \in X$: $\rho(x, y) = \rho(y, x)$. There also must be a pair $x, y \in X$ for which $\rho(x, y) > 0$.

A symmetric kernel ρ is called c -approximate metric if in addition for every $x, y, z \in X$. $\rho(x, z) \leq c(\rho(x, y) + \rho(y, z))$, and $\rho(x, x) = 0$.

For example, if $\rho = \nu^p$, and ν is a metric then ρ is 2^{p-1} -approximate metric .

Definition 2. An (n, d) -graph $G = (V, E)$ is said to satisfy the Poincaré inequality for a symmetric kernel ρ on X with constant γ if for every $f : V \rightarrow X$,

$$\mathbb{E}_{x, y \in V} \rho(f(x), f(y)) \leq \gamma \mathbb{E}_{(x, y) \in E} \rho(f(x), f(y)). \quad (1)$$

The smallest (infimum) such γ is denoted $\gamma(G, \rho)$.

We will heavily use the double cover variant of the above definition.

Definition 3. An (n, d) -graph $G = (V, E)$ is said to satisfy the bipartite Poincaré inequality for a symmetric kernel ρ on X with constant $\bar{\gamma}$ if for every $f, g : V \rightarrow X$,

$$\mathbb{E}_{x, y \in V} \rho(f(x), g(y)) \leq \bar{\gamma} \mathbb{E}_{(x, y) \in E} \rho(f(x), g(y)). \quad (2)$$

The smallest (infimum) such $\bar{\gamma}$ is denoted $\bar{\gamma}(G, \rho)$.

Remark 1. In (2) it is important that both (x, y) and (y, x) appear as distinct pairs in E .

2.2. Examples.

Proposition 2.1. Let $G = (V, E)$ be an (n, d) graph. Then for every $f : V \rightarrow \mathbb{R}$,

$$\mathbb{E}_{x, y \in V} |f(x) - f(y)|^2 \leq \frac{1}{1 - \lambda_2} \mathbb{E}_{(x, y) \in E} |f(x) - f(y)|^2,$$

and this is tight, i.e., $\gamma(G, |\cdot|^2) = \frac{1}{1 - \lambda_2}$.

Proof. Without loss of generality, we may assume that $\mathbb{E} f = 0$, So

$$\mathbb{E}_{x, y \in V} |f(x) - f(y)|^2 = 2 \mathbb{E}_{x \in V} f(x)^2.$$

On the other hand, let A be the doubly stochastic transition matrix of the random walk associated with G . Then

$$\sum_{x, y \in V} (I - A)_{xy} f(x) f(y) = \sum_V f(x)^2 - \frac{1}{d} \sum_{(x, y) \in E} f(x) f(y) = \frac{1}{2d} \sum_{(x, y) \in E} (f(x) - f(y))^2.$$

Write $f = \sum_i a_i v_i$, where $v_i : V \rightarrow \mathbb{R}$ are the orthonormal eigenvectors basis of A (and hence of $I - A$). Note that $v_1 = (n^{-1/2}, \dots, n^{-1/2})^T$, and therefore $0 = \mathbb{E} f = n^{-1/2} \langle f, v_1 \rangle = a_1 / \sqrt{n}$. Hence $a_1 = 0$, and we

have

$$\begin{aligned}
\frac{1}{2d} \sum_{(x,y) \in E} (f(x) - f(y))^2 &= \sum_{x,y \in V} (I - A)_{xy} \left(\sum_i a_i v_i(x) \right) \left(\sum_j a_j v_j(y) \right) \\
&= \sum_{i,j} a_i a_j \sum_{x \in V} v_i(x) \sum_{y \in V} (I - A)_{xy} v_j(y) \\
&= \sum_{i,j} a_i a_j \sum_{x \in V} v_i(x) (1 - \lambda_j) v_j(x) \\
&= \sum_{i,j} a_i a_j (1 - \lambda_j) \sum_{x \in V} v_i(x) v_j(x) \\
&= \sum_i a_i^2 (1 - \lambda_i) \sum_x v_i(x)^2 \\
&\geq (1 - \lambda_2) \sum_{x \in V} \sum_i (a_i v_i(x))^2 \\
&= (1 - \lambda_2) \sum_{x \in V} f(x)^2.
\end{aligned}$$

The tightness is checked by taking $f(x) = v_2(x)$. □

Proposition 2.2. *Let $G = (V, E)$ be an (n, d) , then for every $f, g : V \rightarrow \mathbb{R}$,*

$$\mathbb{E}_{x,y \in V} |f(x) - g(y)|^2 \leq \frac{1}{1 - \lambda} \mathbb{E}_{(x,y) \in E} |f(x) - g(y)|^2,$$

and this is tight, i.e., $\bar{\gamma}(G, |\cdot|^2) = \frac{1}{1 - \lambda}$.

Proof. Without loss of generality, we may assume that $\mathbb{E} f = 0$. So

$$\mathbb{E}_{x,y \in V} |f(x) - g(y)|^2 = \mathbb{E}_{x \in V} f(x)^2 + \mathbb{E}_{x \in V} g(x)^2.$$

On the other hand, let A be the doubly stochastic transition matrix of the random walk associated with G . Then

$$\begin{aligned}
\frac{1}{nd} \sum_{(x,y) \in E} (f(x) - g(y))^2 &= \frac{1}{n} \sum_{(x,y) \in V} A_{xy} (f(x) - g(y))^2 \\
&= \frac{1}{n} \sum_{x \in V} (f(x)^2 + g(x)^2) - \frac{2}{n} \sum_{x,y \in V} A_{xy} f(x) g(y).
\end{aligned}$$

Writing $f = \sum_i a_i v_i$, $g = \sum_i b_i v_i$ where $v_i : V \rightarrow \mathbb{R}$ are the orthonormal eigenvectors basis of A (and hence, of $I - A$), and $a_1 = 0$, since $\mathbb{E} f = 0$. We have

$$\begin{aligned}
\sum_{x,y \in V} A_{xy} f(x) g(y) &= \sum_{x,y \in V} A_{xy} \left(\sum_i a_i v_i(x) \right) \left(\sum_j b_j v_j(y) \right) \\
&= \sum_{i,j} a_i b_j \sum_{y \in V} v_j(y) \sum_{x \in V} A_{xy} v_i(x) = \sum_{i \geq 2, j} a_i b_j \sum_{y \in V} v_j(y) \lambda_i v_i(y) = \sum_{i \geq 2} a_i b_i \lambda_i \sum_{y \in V} v_i(y)^2 \\
&\leq \lambda \sum_i (a_i^2 + b_i^2) / 2 \sum_{y \in V} v_i(y)^2 \leq \frac{\lambda}{2} \sum_{y \in V} (f(y)^2 + g(y)^2).
\end{aligned}$$

The tightness is checked by considering two cases: Either $f(x) = g(x) = v_2(x)$, or $f(x) = -g(x) = v_n(x)$. □

Proposition 2.3. *Let $G = (V, E)$ be an (n, d) , $\gamma(G \otimes \tau, |\cdot|^2) = \frac{1}{1 - \lambda(G)}$.*

Proof. From Proposition 2.1, we know that $\gamma(G \otimes \tau, |\cdot|^2) = \frac{1}{1 - \lambda_2(G \otimes \tau)}$. Hence it remains to compute the spectrum of $G \otimes \tau$, but a straight forward confirmation reveals that $\{\lambda_i(G \otimes \tau)\}_i = \{\pm \lambda_i(G)\}_i$, and hence $\lambda_2(G \otimes \tau) = \max\{\lambda_2(G), -\lambda_n(G)\} = \lambda(G)$. □

Proposition 2.4. Let $G = (V, E)$ be an (n, d) , then for every $f : V \rightarrow \mathbb{R}$,

$$\mathbb{E}_{x, y \in V} |f(x) - f(y)| \leq \frac{2d}{nh} \mathbb{E}_{(x, y) \in E} |f(x) - f(y)|,$$

where

$$h = \min_{\emptyset \neq S \subset V} \frac{|E(S, V \setminus S)|}{|S|(n - |S|)}$$

is the conductance (which is 2 approximation of the Cheeger constant)

Proposition 2.5. Let $G = (V, E)$ be an (n, d) , then for every $f, g : V \rightarrow \mathbb{R}$,

$$\mathbb{E}_{x, y \in V} |f(x) - g(y)| \leq \frac{2d}{n\bar{h}} \frac{n}{d} \mathbb{E}_{(x, y) \in E} |f(x) - g(y)|,$$

where

$$\bar{h} = \min_{\emptyset \neq S, T \subset V} \frac{|E(S, V \setminus T)| + |E(T, V \setminus S)|}{|S|(n - |T|) + |T|(n - |S|)}.$$

Proposition 2.6. Let C_n denote the cycle on n vertices. For any c -approximate metric space (X, ρ) , and any $n \in \mathbb{N}$, $\gamma(C_n, \rho) \leq c^{1+2 \log n}$. Furthermore, if n is odd then $\bar{\gamma}(C_n, \rho) \leq 2c^{2 \log n}$.

Proof. Assume that n is odd. Denote the vertices of C_n by the numbers $\{0, \dots, n-1\}$. Fix $f, g : \{0, \dots, n-1\} \rightarrow X$, and fix $0 \leq i \leq j \leq n-1$. If $j-i$ is odd, then by the c -approximate triangle inequality

$$\rho(f(i), f(j)) \leq c^{2 \log n} \sum_{k=0}^{(j-i-1)/2} (\rho(f(i+2k), g(i+2k+1)) + \rho(f(i+2k+2), g(i+2k+1))).$$

Similarly, if $j-i$ is even, the $n-(j-i)$ is odd and we can write

$$\begin{aligned} \rho(f(i), f(j)) \leq c^{2 \log n} \sum_{k=0}^{(n-(j-i)-1)/2} & \left(\rho(f(i+2k \pmod n), g(i+2k+1 \pmod n)) \right. \\ & \left. + \rho(f(i+2k+2 \pmod n), g(i+2k+1 \pmod n)) \right). \end{aligned}$$

Hence by summing over all $i, j \in \{0, \dots, n-1\}$, and using obvious symmetries we conclude that

$$\mathbb{E}_{i, j \in C_n} \rho(f(i), g(j)) \leq c^{1+2 \log n} \mathbb{E}_{\substack{i, j \in C_n \\ |i-j|=1}} \rho(f(i), g(j)).$$

The bound on $\gamma(C_n, \rho)$ when n is even is similar (and slightly simpler). $\square$

Similarly, we can prove

Proposition 2.7. Let C_n^o denote the cycle on n vertices, with self loops. For any c -approximate metric space (X, ρ) , and any $n \in \mathbb{N}$, $\bar{\gamma}(C_n^o, \rho) \leq 2c^{1+2 \log n}$. $\square$

2.3. Relating γ and $\bar{\gamma}$.

Proposition 2.8. For every kernel space (X, ρ) , and G , $\gamma(G, \rho) \leq \bar{\gamma}(G, \rho)$.

Proof. For every $f : V \rightarrow X$, write $g := f$, and from the definition,

$$\mathbb{E}_{x, y \in V} \rho(f(x), f(y)) \leq \bar{\gamma} \mathbb{E}_{(x, y) \in E} \rho(f(x), f(y)). \quad \square$$

Lemma 2.9. Assume $G = (V_1, V_2, E)$ is a bipartite $(2n, d)$ graph with $|V_1| = |V_2| = n$, and ρ is a kernel over X . Then there exists a $(n, 4d)$ graph $G' = (V', E')$ such that $\bar{\gamma}(G', \rho) \leq 2\gamma(G, \rho)$.

Proof. Fix an arbitrary bijection $m : V_1 \rightarrow V_2$. Denote $V = V_1 \cup V_2$, and $\widehat{E} = E \cup \{(m(u), m(v)) : (u, v) \in E\}$. Fix $f : V \rightarrow X$. First note that

$$\mathbb{E}_{\substack{u \in V_1 \\ v \in V_2}} \rho(f(u), f(v)) \leq 2 \mathbb{E}_{u, v \in V} \rho(f(u), f(v)) \leq 2\gamma(G) \mathbb{E}_{(u, v) \in E} \rho(f(u), f(v)).$$

Similarly

$$\mathbb{E}_{u,v \in V} \rho(f(u), f(v)) = \mathbb{E}_{u,v \in V} \rho(f(m(u)), f(m(v))) \leq \gamma(G) \mathbb{E}_{(u,v) \in E} \rho(f(m(u)), f(m(v)))$$

So, by averaging the previous two inequalities, we obtain

$$\mathbb{E}_{\substack{u \in V_1 \\ v \in V_2}} \rho(f(u), f(v)) \leq 2 \mathbb{E}_{u,v \in V} \rho(f(u), f(v)) \leq 2\gamma(G) \mathbb{E}_{(u,v) \in \hat{E}} \rho(f(u), f(v)).$$

We have just proved that for the graph $G' = (V_1, \{(u, v) : u, v \in V_1, (u, m(v)) \in \hat{E}\})$, $\bar{\gamma}(G') \leq 2\gamma(G)$. $\square$

Proposition 2.10. *Assume (X, ρ) is c -approximate metric space, and $G = (V, E)$ is a connected (n, d) graph with even n . Then there exists a $(n/2, 16d)$ graph $G' = (V', E')$ for which $\bar{\gamma}(G') \leq 8c\gamma(G)$.*

Proof. Arbitrarily, split $V = V_1 \cup V_2$ to two subsets of the same size, and fix an arbitrary bijection $m : V_1 \rightarrow V_2$. Denote by $m(v)$ the vertex matched to v . Define the edge multi-set

$$\begin{aligned} \hat{E} = & \{(u, m(v)) : u, v \in V_1, (u, v) \in E\} \\ & \cup \{(u, m(v)) : u, v \in V_1, (m(u), m(v)) \in E\} \\ & \cup \{(u, m(v)) : u, v \in V_1, (u, m(v)) \in E\} \\ & \cup \{(u, m(v)) : u, v \in V_1, (m(u), v) \in E\} \\ & \cup \{2d \times (u, m(u)) : u \in V_1\}. \end{aligned}$$

Clearly, $\hat{G} = (V_1, V_2, \hat{E})$ is a bipartite $(n, 4d)$ graph.

With every edge $(u, v) \in E \cap V_1 \times V_1$, we can associate two unique pairs of edges in $\hat{E}$: The pair $(u, m(v)), (v, m(v)) \in \hat{E}$, and the pair $(v, m(u)), (u, m(u)) \in \hat{E}$. For the first pair we have $\rho(u, v) \leq c \cdot \rho(u, m(v)) + c \cdot \rho(v, m(v))$, and similarly with the second pair. Since we have put $2d$ copies of the edge $(v, m(v))$ in $\hat{E}$, no edge is “used twice” in this matching, although some of them may not be used at all.

We can now bound $\gamma(\hat{G}, \rho)$.

$$\begin{aligned} \mathbb{E}_{V \times V} \rho(f(u), f(v)) & \leq \frac{\gamma(G, \rho)}{nd} \cdot \sum_E \rho(f(u), f(v)) \\ & \leq \frac{c\gamma(G, \rho)}{nd} \cdot \sum_{\hat{E}} \rho(f(u), f(v)) \\ & = 4c\gamma(G, \rho) \cdot \mathbb{E}_{\hat{E}} \rho(f(u), f(v)). \end{aligned}$$

Hence $\gamma(\hat{G}, \rho) \leq 4c\gamma(G, \rho)$. Since $\hat{G}$ is bipartite, applying Proposition 2.9, we conclude that there exists a $(n/2, 8d)$ graph G' for which $\bar{\gamma}(G', \rho) \leq 2\gamma(\hat{G}, \rho) \leq 8c\gamma(G, \rho)$. $\llcorner$ No need to lose a factor of 2 in the degree. $\square$ MM
—Manor $\gg$

Lemma 2.11. *For any c -approximate metric space (X, ρ) ,*

$$\frac{1}{c+0.5} \cdot \gamma(G \otimes \tau, \rho) \leq \bar{\gamma}(G, \rho) \leq 2\gamma(G \otimes \tau, \rho). \quad (3)$$

Proof. We begin with the upper bound in (3). Fix $f, g : V \rightarrow X$, and define $h : V \times \{0, 1\} \rightarrow X$,

$$h(x, a) = \begin{cases} f(x) & a = 0 \\ g(x) & a = 1. \end{cases}$$

Then

$$\begin{aligned} \mathbb{E}_{x,y \in V} \rho(f(x), g(y)) & = \mathbb{E}_{x,y \in V} \rho(h(x, 0), h(y, 1)) \\ & \leq 2 \mathbb{E}_{\substack{x,y \in V \\ a,b \in \{0,1\}}} \rho(h(x, a), h(y, b)) \\ & \leq 2\gamma(G \otimes \tau, \rho) \mathbb{E}_{(x,y) \in E} \rho(h(x, 0), h(y, 1)). \end{aligned}$$

We next prove the lower bound in (3). Fix $h : V \times \{0, 1\} \rightarrow X$, and define $f(x) = h(x, 0)$, and $g(x) = h(x, 1)$. For $a \in \{0, 1\}$

$$\begin{aligned} & \mathbb{E}_{x, y \in V} \rho(h(x, a), h(y, a)) \\ & \leq \mathbb{E}_{x, y \in V} c \mathbb{E}_{z \in V} (\rho(h(x, a), h(z, 1 - a)) + \rho(h(y, a), h(z, 1 - a))) = 2c \mathbb{E}_{x, y \in V} \rho(f(x), g(y)). \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}_{\substack{x, y \in V \\ a, b \in \{0, 1\}}} \rho(h(x, a), h(y, b)) &= \frac{1}{2} \cdot \mathbb{E}_{\substack{x, y \in V \\ a \in \{0, 1\}}} \rho(h(x, a), h(y, a)) + \frac{1}{2} \cdot \mathbb{E}_{x, y \in V} \rho(h(x, 0), h(y, 1)) \\ &\leq (c + 0.5) \cdot \mathbb{E}_{x, y \in V} \rho(f(x), g(y)) \\ &\leq (c + 0.5) \cdot \bar{\gamma}(G, \rho) \mathbb{E}_{(x, y) \in E} \rho(h(x, 0), h(y, 1)). \end{aligned} \quad \square$$

We will also use the following simple fact.

Lemma 2.12. *Fix a symmetric and reflexive kernel ρ on X (i.e., it also satisfies $\rho(x, x) = 0$), let $G = (V, E)$, be an (n, d) graph, and let $d' \geq d$ satisfying either one of the following conditions:*

- d' is a multiple of d or
- both d' and d are even.

Then there exists an (n, d') graph $G' = (V, E')$ such that

$$\gamma(G', \rho) \leq 2\gamma(G, \rho) \quad \text{and} \quad \bar{\gamma}(G', \rho) \leq 2\bar{\gamma}(G, \rho).$$

Furthermore, if G is a Cayley graph of a group Γ , then so does G' .

Proof. First observe that if d' is a multiple of d then we can simply duplicating every edge of G , d'/d times, which does not change γ and $\bar{\gamma}$ at all. To reach d' which is not a multiple of d , we first apply the duplication process to reach degree $h = \lfloor d'/d \rfloor \cdot d$, and then we add $(d' - h)/2$ self loops for every vertex. Note that in Cayley graph both operations amounts for just adding generators. Also note that the number of self loop edges amount to at most half the total number of edges, and hence the average over the edges decrease by at most a factor of two. $\square$

We call the operation performed in Lemma 2.12, d' -edge completion of G and it is denoted by $G' = \mathcal{C}_{d'}(G)$.

3. BASE GRAPH

The purpose of this section is mainly to prove.

Theorem 1. *Let $(X, \|\cdot\|)$ be a K -convex Banach space, and $p > 1$. Then there exist $\delta = \delta(X, p) > 0$, $C = C(X, p) > 0$ and a family of unbounded size Abelian Cayley graphs $\{G_i\}_{i \in \mathbb{N}}$ such that G_i is (N_i, d_i) graph with $d_i \leq \exp(\log^{1-\delta} N_i)$ and $\gamma(G_i, \|\cdot\|^p) \leq C$.*

3.1. Notation and basic facts. Denote $[n] = \{1, \dots, n\}$ Let X be a K -convex Banach space. Let $f : \{\pm 1\}^n \rightarrow X$. Define the partial derivative as

$$\partial_i f = \frac{f(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_n)}{2},$$

and the Laplacian is

$$\Delta f = \sum_{i=1}^n \partial_i f$$

Let $\{W_A\}_{A \subseteq [n]}$ be the Walsh basis of $\mathbb{R}^{\{\pm 1\}^n}$, i.e.,

$$W_A(x_1, \dots, x_n) = \prod_{i \in A} x_i.$$

Denote by $L_p(X) = L_p(X, \{\pm 1\}^n, \mu)$, where μ is the uniform measure on $\{\pm 1\}^n$. Every $f \in L_p(X)$ can be written as $\sum_{A \subset [n]} \hat{f}(A) W_A$, where $\hat{f}(A) \in X$. For $f \in L_p(X)$, we write

$$\|f\|_p = \|f\|_{L_p(X)} = \left(\sum_{a \in \{\pm 1\}^n} 2^{-n} \|f(a)\|_X^p \right)^{1/p}.$$

Note that

$$\partial_i W_A = \begin{cases} W_A & i \in A \\ 0 & i \notin A \end{cases},$$

and so $\partial_i f = \sum_{A: i \in A} \hat{f}(A) W_A$. Similarly $\Delta f = \sum_A |A| \hat{f}(A) W_A$.

We also define the k th level Rademacher projection by

$$R_k f = \sum_{A: |A|=k} \hat{f}(A) W_A.$$

Let us also define

$$L_p^{\geq m}(X) = L_p^{\geq m}(X, \{\pm 1\}^n, \mu) = \{f \in L_p(X) : \forall A, |A| < m \Rightarrow \hat{f}(A) = 0\}.$$

The Beckner operator on $L_p(X)$ is defined as

$$(T_t f)(x) := \sum_{A \subset [n]} e^{-t|A|} \hat{f}(A) W_A(x) = \sum_{y \in \{\pm 1\}^n} \left(\frac{1-e^{-t}}{2} \right)^{\|x-y\|_1} \left(\frac{1+e^{-t}}{2} \right)^{n-\|x-y\|_1} f(y). \quad (4)$$

The second equality easily follows by computing the third term in (4) when $f = W_A$. We write $x = x_A x_{A^c}$, and

$$\begin{aligned} & \sum_{y \in \{\pm 1\}^n} \left(\frac{1-e^{-t}}{2} \right)^{\|x-y\|_1} \left(\frac{1+e^{-t}}{2} \right)^{n-\|x-y\|_1} \prod_{i \in A} y_i \\ &= \sum_{y_A \in \{\pm 1\}^A} \left(\frac{1-e^{-t}}{2} \right)^{\|x_A - y_A\|_1} \left(\frac{1+e^{-t}}{2} \right)^{|A| - \|x_A - y_A\|_1} \prod_{i \in A} y_i \\ & \quad \cdot \sum_{y_{A^c} \in \{\pm 1\}^{[n] \setminus A}} \left(\frac{1-e^{-t}}{2} \right)^{\|x_{A^c} - y_{A^c}\|_1} \left(\frac{1+e^{-t}}{2} \right)^{(n-|A|) - \|x_{A^c} - y_{A^c}\|_1} \\ &= \sum_{\ell=0}^{|A|} \binom{|A|}{\ell} \left(\frac{1-e^{-t}}{2} \right)^\ell \cdot \left(\frac{1+e^{-t}}{2} \right)^{|A|-\ell} (-1)^\ell W_A(x) \\ &= e^{-t|A|} W_A(x). \end{aligned}$$

For $x \in \{\pm 1\}^n$, and $\tau \in [0, 1]$, we also denote by $\mathcal{N}(x, \tau)$ the probability distribution on $\{\pm 1\}^n$ such that $\Pr[\mathcal{N}(x, \tau) = y] = \tau^{\|x-y\|_1} (1-\tau)^{n-\|x-y\|_1}$. So

$$(T_t f)(x) = \mathbb{E}_{y \sim \mathcal{N}(x, (1-e^{-t})/2)} f(y).$$

3.2. Bounding the Beckner Operator on the tail.

Lemma 3.1. *For every K -convex Banach Space X , and $p > 1$, there exist $\alpha, a > 0$, such that for every $nm \in \mathbb{N}$, $m \leq n$, and $t \geq 0$.*

$$\|T_t\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} \leq c e^{-am \min\{t, t^\alpha\}}, \quad (5)$$

Proof. First complexify $X \ll$ HOW TO DO THAT? Do you define $\|x + iy\| = (\|x\|_X^2 + \|y\|_X^2)^{1/2}$? —Manor>> MM such that its K -convex constant does not change. Next extend the Beckner operator $T_z : L_p(X) \rightarrow L_p(X)$ to $z \in \mathbb{C}$, as

$$T_z f = \sum_{A \subset [n]} \exp(-z|A|) \hat{f}(A) W_A.$$

Another way to write it is $T_z = \sum_{k=0}^n e^{-zk} R_k$.

Theorem 2 (Pisier). *For K -convex X , and $p \in (1, \infty)$, there exists $\phi, M > 0$ such that if $|\arg z| < \phi$ then $\|T_z\|_{L_p(X) \rightarrow L_p(X)} \leq M$*

Corollary 3.2. *Let There exists $M, a > 0$ such that $\|R_k\|_{L_p(X) \rightarrow L_p(X)} \leq Me^{ak}$*

Proof. Define $a = \pi / \tan \phi$, so that the segment $\{(a, iy) : y \in [-\pi, \pi]\}$ is all with argument at most ϕ . Calculate

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ikt} T_{a+it} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ikt} \sum_{r=0}^n e^{-(a+it)r} R_r dt = \frac{1}{2\pi} \sum_{r=0}^n e^{-ra} R_r \int_{-\pi}^{\pi} e^{i(k-r)t} dt = e^{-ka} R_k,$$

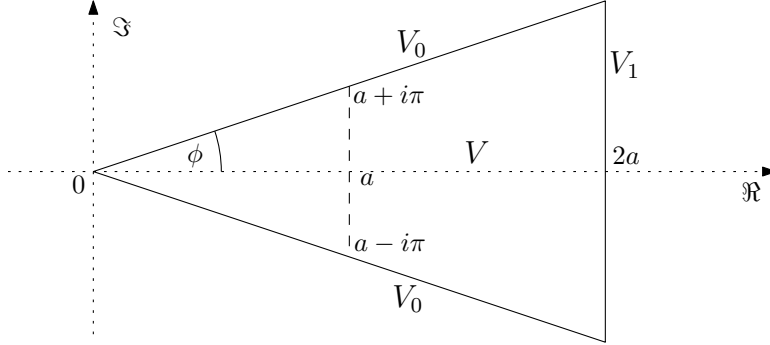
and therefore, by convexity,

$$\|R_k\|_{L_p(X) \rightarrow L_p(X)} \leq \frac{e^{ka}}{2\pi} \int_{-\pi}^{\pi} |e^{ikt}| \cdot \|T_{a+it}\|_{L_p(X) \rightarrow L_p(X)} dt \leq Me^{ka}. \quad \square$$

Corollary 3.3. *Assume that $\Re z \geq 2a$ then*

$$\|T_z\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} = \left\| \sum_{k \geq m} e^{-zk} R_k \right\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} \leq \sum_{k \geq m} e^{-2ak} \cdot Me^{ak} \leq Ce^{-ma}. \quad \square$$

Next, consider the triangle $V \subset \mathbb{C}$ define by the vertices 0 , $2a + i2\pi$, and $2a - i2\pi$. Denote by $V_1 = \{2a + iy : y \in [-2\pi, 2\pi]\}$, and $V_0 = \{x \pm ix \tan \phi : x \in [0, 2a)\}$, so $\partial V = V_0 \cup V_1$.



Fix $t \in (0, 2a)$

Lemma 3.4 (Harmonic measure). *There exists a probability measure $\mu = \mu_t$ on ∂V such that for every analytic function on V ,*

$$f(t) = \int_{\partial V} f(z) d\mu(z).$$

Let $\theta = \mu(V_1)$, and denote $\mu = (1 - \theta)\mu_0 + \theta\mu_1$, where μ_i is a probability measure on V_i , $i \in \{0, 1\}$.

Lemma 3.5. $\theta \sim_{\phi, a} t^{\pi/2\phi}$. $\square$

Lemma 3.6. *For every $\varepsilon > 0$ there exists an analytic function on V $\Psi : V \rightarrow \mathbb{C}$, such that $\Psi(t) = 1$, for every $z \in V_0$, $|\Psi(z)| \leq \varepsilon$, and for every $z \in V_1$, $|\Psi(z)| \leq \varepsilon^{-\theta^{-1}+1}$*

We next calculate

$$T_t = \Psi(t)T_t = \int_{\partial V} \Psi(z)T_z d\mu(z) = (1 - \theta) \int_{V_0} \Psi(z)T_z d\mu_0(z) + \theta \int_{V_1} \Psi(z)T_z d\mu_1(z).$$

Hence

$$\|T_t\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} \leq (1 - \theta)\varepsilon M + \theta\varepsilon^{1-\theta^{-1}}Ce^{-am}.$$

Since this inequality is true for every $\varepsilon > 0$, we can minimize it over $\varepsilon > 0$. Consider

$$g(\varepsilon) = (1 - \theta)\varepsilon M + \theta\varepsilon^{1-\theta^{-1}}Ce^{-am}.$$

$$g'(\varepsilon) = (1 - \theta)M + \theta(1 - \theta^{-1})\varepsilon^{-\theta^{-1}}Ce^{-am} = 0$$

g is minimized when $\varepsilon_0 \sim e^{-\theta ma}$, which means that $g(\varepsilon_0) \sim e^{-am\theta}$. Hence, for $t \in (0, 2a)$,

$$\|T_t\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} \leq ce^{-amt^\alpha},$$

where $\alpha \approx \pi/2\phi$. All in all, for $t \in [0, \infty)$,

$$\|T_t\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} \leq ce^{-am \min\{t^\alpha, t\}}. \quad \square$$

As a side note we show a corollary that give an interesting lower bound on the Laplacian in the tail space:

Theorem 3. *If X is a K -convex Banach space then there exist $\varepsilon \in (0, 1)$, and $c > 0$ such that*

$$\|\Delta f\|_{L_p^{\geq m}(X)} \geq cm^\varepsilon \|f\|_{L_p(X)}.$$

Proof. When $m = 0$ the lower bound is vacuous. Observe that on $L_p^{\geq 1}$, Δ is invertible, and $\Delta^{-1}f = \sum_A |A|^{-1} \hat{f}(A) W_A$. Further note that

$$\left(\int_0^\infty T_t dt\right) W_A = \int_0^\infty T_t W_A dt = \int_0^\infty e^{-t|A|} W_A dt = |A|^{-1} W_A,$$

so $\Delta^{-1} = \int_0^\infty T_t dt$. Therefore

$$\|\Delta^{-1}\|_{L_p^{\geq m}(X) \rightarrow L_p^{\geq m}(X)} \leq \int_0^\infty \|T_t\| dt \leq \int_0^\infty ce^{-am \min\{t^\alpha, t\}} dt \lesssim m^{-1/\alpha}. \quad \square$$

Corollary 3.7. *Let X be K -convex space, and $p > 1$. Then there exist $C, \delta > 0$ such that for a linear code with minimum distance m , $C \subset \mathbb{F}_2^n$ we have $\gamma(\mathbb{F}_2^n/C^\perp, \|\cdot\|_X^p) \leq cn^p m^{-\delta}$.*

Proof. Think of f as $f : \mathbb{F}_2^n \rightarrow X$ being constant on $x + C^\perp$, as argued in [13, Lemma 3.3], this means that $\hat{f}(A) = 0$ for every $A \subset [n]$, for which $|A| < m$. Hence by Theorem 3,

$$\begin{aligned} \int_{\mathbb{F}_2^n \times \mathbb{F}_2^n} \|f(x) - f(y)\|_X^p d\mu(x) d\mu(y) &\leq 2^p \int_{\mathbb{F}_2^n} \|f(x)\|_X^p = 2^p \|f\|_p^p \leq (2cm^{-\varepsilon})^p \|\Delta f\|_p^p \\ &= Cm^{-\delta} \left\| \sum_i \partial_i f \right\|_p^p \leq Cn^{p-1} m^{-\delta} \sum_i \|\partial_i f\|_p^p. \quad \square \end{aligned}$$

The bound in Corollary 3.7 on $\gamma(\mathbb{F}_2^n/C^\perp, \|\cdot\|_X^p)$ is non-trivial, but unfortunately it is not good enough for constructing zigzag expanders. To overcome this problem, in the next section we will replace the standard edges of $\mathbb{F}_2^n$ with “noisy edges”.

3.3. Quotient of Noisy Cube. Fix $C \subset \mathbb{F}_2^n$ a linear subspace of dimension at most $0.9n$ and with minimum distance $m \geq 0.1n$, and $f : \mathbb{F}_2^n/C^\perp \rightarrow X$. Identify f with $f : \mathbb{F}_2^n \rightarrow X$ which is constant on cosets of $C^\perp$. As observed by Khot and Naor, $\hat{f}(A) = 0$ for every $|A| < m$, so applying (5) with $t \in [0, 1]$, we bound

$$\|T_t f\|_{L_p(X)} \leq ce^{-amt^\alpha} \|f\|_{L_p(X)},$$

for some $\alpha \geq 1$. Using the triangle inequality

$$\|(I - T_t)f\|_{L_p(X)} \geq \|f\|_{L_p(X)} - \|T_t f\|_{L_p(X)} \geq (1 - ce^{-amt^\alpha}) \|f\|_{L_p(X)}.$$

So, denoting $\tau = (1 - e^{-t})/2$,

$$\begin{aligned} \mathbb{E}_{x, y \in \{\pm 1\}^n} \|f(x) - f(y)\|_X^p &\leq 2^p \|f\|_{L_p(X)}^p \\ &\leq 2^p (1 - ce^{-amt^\alpha})^{-p} \|(I - T_t)f\|_{L_p(X)}^p \\ &= 2^p (1 - ce^{-amt^\alpha})^{-p} 2^{-n} \sum_x \left\| \sum_y \tau^{\|x-y\|_1} (1 - \tau)^{n-\|x-y\|_1} (f(x) - f(y)) \right\|_X^p \\ &\leq 2^p (1 - ce^{-amt^\alpha})^{-p} 2^{-n} \sum_x \sum_y \tau^{\|x-y\|_1} (1 - \tau)^{n-\|x-y\|_1} \|f(x) - f(y)\|_X^p \\ &= 2^p (1 - ce^{-amt^\alpha})^{-p} \mathbb{E}_{x \in \{\pm 1\}^n} \mathbb{E}_{y \sim \mathcal{N}(x, \tau)} \|f(x) - f(y)\|_X^p. \end{aligned} \quad (6)$$

This is similar to a Poincaré inequality, when $t \leq n^{-1/\alpha}$, and the multiplicative factor on the right hand side becomes a constant depending on X but not on n . We choose $t = n^{-\min\{\alpha^{-1}, 0.3\}}$. this means that $\tau = (1 - e^{-t})/2 \sim n^{-\min\{\alpha^{-1}, 0.3\}}$.

The main problem left is that the degree in the relevant graph (whose vertex set is $\mathbb{F}_2^n/C^\perp$) is much too large. Our solution is to approximate the noise distribution using small degree unweighted graph by truncating to zero pairs having weight below some threshold, and approximate the rest using multiple parallel edges.

«Need to make consistent references to either $\mathbb{F}_2^n$ or $\{\pm 1\}^n$ —Manor»

Lemma 3.8. *There exists a Cayley multi-graph on $\mathbb{F}_2^n$, $G = (\mathbb{F}_2^n, E)$, having degree at most $\tau^{-4\tau n}(1 - \tau)^{-(1-4\tau)n}$ such that for every c -approximate metric space (X, ρ) and $f : \mathbb{F}_2^n \rightarrow X$,*

$$\frac{1}{3} \mathbb{E}_{(x,y) \in E} \rho(f(x), f(y)) \leq \mathbb{E}_{x \in \{\pm 1\}^n} \mathbb{E}_{y \sim \mathcal{N}(x, \tau)} \rho(f(x), f(y)) \leq 3 \mathbb{E}_{(x,y) \in E} \rho(f(x), f(y)), \quad (7)$$

as long as $18\tau^2 n \geq 2 \log c \cdot \log n + \log 4$.

Proof. The edges from 0 are defined as follows (the rest of the edges are defined by translation). Fix $\sigma = \tau^{4\tau n}(1 - \tau)^{(1-4\tau)n}$. We connect y to 0 with $\lfloor \tau^{\|y\|_1}(1 - \tau)^{n-\|y\|_1}/\sigma \rfloor$ edges.

Denote by $\deg_E(0)$ the number of edges leaving 0.

Claim 1. $\sigma^{-1}/3 \leq \deg_E(0) \leq \sigma^{-1}$

Proof. The upper bound is obvious.

Denote by $|(0, y)|$ the number of edges between 0 and y . Note that $|(0, y)| = 0$ if and only if $\|y\|_1 > 4\tau n$, and By Chernoff inequality «cf. [3, Thm A.1.4], but should change to binomial approximation —Manor» $\sum_{\ell \geq 4\tau n} \binom{n}{\ell} \tau^\ell (1 - \tau)^{n-\ell} \leq \exp(-18\tau^2 n)$. For $\|y\|_1 \leq 4\tau n$, $|(0, y)| \geq \tau^{\|y\|_1}(1 - \tau)^{n-\|y\|_1}/2\sigma$, and therefore $\deg_E(0) \geq (1 - \exp(-18\tau^2 n))\sigma^{-1}/2 \geq \sigma^{-1}/3$. $\square$

From the above claim it is clear that

$$\mathbb{E}_{(0,y) \in E} \rho(f(0), f(y)) = \frac{\sigma^{-1}}{\deg_E(0)} \cdot \frac{\sum_{(0,y) \in E} \rho(f(0), f(y))}{\sigma^{-1}} \leq 3 \mathbb{E}_{y \sim \mathcal{N}(0, \rho)} \rho(f(0), f(y)),$$

which proves the lower bound in (7).

The upper bound in (7) is more cumbersome to prove and requires the use of the approximate triangle inequality property of ρ .

Fix an even length $s > 4\tau n$, and a length $t \leq 4\tau n$. If $y \in \mathbb{F}_2^n$, $\|y\|_1 = s$, then we can write $0 = y_0, y_1, \dots, y_\ell = y$, where $\ell \leq n/2$ and $\|y_i - y_{i-1}\|_1 = t$. Hence by the c -approximate triangle inequality,

$$\rho(0, y) \leq c^{2 \log n} \cdot \frac{1}{\ell} \sum_{i=0}^{\ell-1} \rho(f(y_i), f(y_{i+1})).$$

Averaging over all possible such paths, and over all pair of source and target $x, y \in \mathbb{F}_2^n$, we obtain from symmetry that

$$\mathbb{E}_{x \in \mathbb{F}_2^n} \mathbb{E}_{\substack{y \in \mathbb{F}_2^n \\ \|y-x\|_1=s}} \rho(f(x), f(y)) \leq c^{2 \log n} \cdot \mathbb{E}_{x \in \mathbb{F}_2^n} \mathbb{E}_{\substack{y \in \mathbb{F}_2^n \\ \|y-x\|_1=t}} \rho(f(x), f(y)) \quad (8)$$

The averages above are uniform. Since the number of edges $|(x, y)|$ between x and y depends only on $\|y - x\|_1$ we can take a weighted average of the right hand side of (8) (concretely: weight of $|(0, y)|/\deg_E(0)$ for distance t , where $\|y\|_1 = t$), and obtain for even $s > 4\tau n$,

$$\mathbb{E}_{x \in \mathbb{F}_2^n} \mathbb{E}_{\substack{y \in \mathbb{F}_2^n \\ \|y-x\|_1=s}} \rho(f(x), f(y)) \leq c^{2 \log n} \cdot \mathbb{E}_{(x,y) \in E} \rho(f(x), f(y)).$$

For odd length s , we can basically do the same but only for odd length t . Fix an odd length $s > 4\tau n$, and an odd length $t \leq 4\tau n$. If $y \in \mathbb{F}_2^n$, $\|y\|_1 = s$, then we can write $0 = y_0, y_1, \dots, y_\ell = y$, where $\ell \leq n/2$ and $\|y_i - y_{i-1}\|_1 = t$. Similar to the even case,

$$\mathbb{E}_{x \in \mathbb{F}_2^n} \mathbb{E}_{\substack{y \in \mathbb{F}_2^n \\ \|y-x\|_1=s}} \rho(f(x), f(y)) \leq c^{2 \log n} \cdot \mathbb{E}_{x \in \mathbb{F}_2^n} \mathbb{E}_{\substack{y \in \mathbb{F}_2^n \\ \|y-x\|_1=t}} \rho(f(x), f(y)). \quad (9)$$

Claim 2. $\Pr_{Y \sim \mathcal{B}(n, \tau)}[Y \text{ is odd}] \geq 1/4$, where $\mathcal{B}$ is the binomial distribution, $\tau < 1/8$, and $\tau n > 50$.

Proof. By Chernoff-type bound (cf. [3, Corollary A.1.14],

$$\Pr[0.5\tau n \leq Y \leq 1.5\tau n] \geq 1 - 2e^{-0.1\tau n} \geq 0.9.$$

When $0.5\tau n \leq k \leq 1.5\tau n$, We bound

$$\frac{\Pr[Y = k+1]}{\Pr[Y = k]} = \frac{\binom{n}{k+1} \tau^{k+1} (1-\tau)^{n-k-1}}{\binom{n}{k} \tau^k (1-\tau)^{n-k}} = \frac{\tau}{1-\tau} \cdot \frac{n-k}{k+1} \in \left[\frac{\tau}{1-\tau} \frac{1-1.5\tau}{1.5\tau}, \frac{\tau}{1-\tau} \frac{1-0.5\tau}{0.5\tau} \right] \subseteq [3^{-1}, 3].$$

The lower bound on the odd numbers is now clear. $\square$

Similarly to the even case, we can apply (9), and then the above claim to obtain

$$\mathbb{E}_{\substack{x \in \mathbb{F}_2^n \\ \|y-x\|_1 = s}} \mathbb{E}_{y \in \mathbb{F}_2^n} \rho(f(x), f(y)) \leq c^{2 \log n} \cdot \mathbb{E}_{\substack{(x,y) \in E \\ \|y-x\|_1 \text{ odd}}} \rho(f(x), f(y)) \leq 4c^{2 \log n} \cdot \mathbb{E}_{(x,y) \in E} \rho(f(x), f(y)).$$

Thus, denote by $\theta = \Pr_{y \sim N(0, \rho)}[\|y\| > 4\tau n] \leq \exp(-18\tau^2 n)$

$$\begin{aligned} & \mathbb{E}_{x \in \{\pm 1\}^n} \mathbb{E}_{y \sim \mathcal{N}(x, \tau)} \rho(f(x), f(y)) \\ &= \theta \mathbb{E}_{x \in \{\pm 1\}^n} \mathbb{E}_{y \sim \mathcal{N}(x, \tau)} [\rho(f(x), f(y)) \mid \|y-x\|_1 > 4\tau n] \\ & \quad + (1-\theta) \mathbb{E}_{x \in \{\pm 1\}^n} \mathbb{E}_{y \sim \mathcal{N}(x, \tau)} [\rho(f(x), f(y)) \mid \|y-x\|_1 \leq 4\tau n] \\ & \leq 4 \exp(-18\tau^2 n) \cdot n^{2 \log c} \cdot \mathbb{E}_{(x,y) \in E} \rho(x, y) + 2 \mathbb{E}_{(x,y) \in E} \rho(x, y) \\ & \leq 3 \mathbb{E}_{(x,y) \in E} \rho(x, y). \end{aligned} \quad \square$$

Proof of Theorem 1. Lets recap. Applying the choice of τ and Lemma 3.8 on (6), we have just found a family of Cayley graphs $(\mathbb{F}_2^n, \hat{E})$ that for sufficiently large n has degree at most $e^{n^{1-\delta}}$, for some $\delta > 0$ independent of n , and satisfies for every $f : \mathbb{F}_2^n \rightarrow X$, which is constant on cosets of $C^\perp$,

$$\mathbb{E}_{x, y \in \{\pm 1\}^n} \|f(x) - f(y)\|_X^p \leq C \mathbb{E}_{(x,y) \in E} \|f(x) - f(y)\|_X^p, \quad (10)$$

where $C > 0$ is independent of n .

To have a honest-to-god Poincaré inequality we should consider the graph $(\mathbb{F}_2^n / C^\perp, \hat{E})$. The problem is that in this graph the bound on the degree is not good enough. The solution in this case is simple. Notice that on the right hand side of (10) we do not care about absolute number of edges, but rather on relative number of edges compared to the other pairs. Since $(\mathbb{F}_2^n, \hat{E})$ is a Cayley graph, every edge $(0, y) \in \hat{E}$ has multiplicity of $|C^\perp|$ in $\mathbb{F}_2^n / C^\perp$ in the form of $(x + C^\perp, x + C^\perp + y)$. Hence we can put $E = \{([x], [x+y]) : [x] \in \mathbb{F}_2^n / C^\perp, (0, y) \in \hat{E}\}$, where here $[x]$ is the coset of x in $\mathbb{F}_2^n / C^\perp$ and is counted only once for all x in the same coset. $\square$

Corollary 3.9. *Let $(X, \|\cdot\|_X)$ be a K -convex Banach space, and $p > 1$. Then there exist $\delta = \delta(X, p) > 0$, $C' = C(X, p) > 0$ and a family of unbounded size graphs $\{G_i\}_{i \in \mathbb{N}}$ such that G_i is (N_i, d_i) graph with $d_i \leq \exp(\log^{1-\delta} N_i)$ and $\bar{\gamma}(G_i, \|\cdot\|_X^p) \leq C$.*

Proof. Apply Proposition 2.10 on the graph from Theorem 1. $\square$

The resulting graph of Corollary 3.9 is not necessarily a Cayley graph, because of Proposition 2.10. However, in ad hoc manner, we can transform the concrete example of quotient of noisy cube into a one with a bound on $\bar{\gamma}$, while keeping the example a Cayley graph.

Corollary 3.10. *Let $(X, \|\cdot\|_X)$ be a K -convex Banach space, and $p > 1$. Then there exist $\delta = \delta(X, p) > 0$, $C' = C(X, p) > 0$ and a family of unbounded size Cayley graphs $\{G_i\}_{i \in \mathbb{N}}$ such that G_i is (N_i, d_i) graph with $d_i \leq \exp(\log^{1-\delta} N_i)$ and $\bar{\gamma}(G_i, \|\cdot\|_X^p) \leq C$. Furthermore, both d_i and N_i are even.*

Sketch of a proof. As before let $C \subset \mathbb{F}_2^n$ a linear subspace of dimension at most $0.9n$ and minimum distance at least $0.1n$. Consider the Linear subspace $D = C \times \{0, 1\} \subset \mathbb{F}_2^{n+1}$. D has similar properties: dimension at most $0.9n + 1$ and minimum distance at least $0.1n$. Note that $D^\perp = C^\perp \times \{0\}$. Hence we have $\mathbb{F}_2^{n+1}/D^\perp = \mathbb{F}_2^n/C^\perp \times \mathbb{F}_2$. Let $(\mathbb{F}_2^{n+1}, F)$ be the graph described in (10) with respect to $f : \mathbb{F}_2^{n+1} \rightarrow X$ which are constant on cosets of $D^\perp$. Assume its degree is d_0 . We next define an edge multiset on $\mathbb{F}_2^n$ by

$$E = \{(u, v) : u, v \in \mathbb{F}_2^n, a, b \in \mathbb{F}_2, (ua, vb) \in F\} \cup \{2d_0 \times (u, u) : u \in \mathbb{F}_2^n\}. \quad (11)$$

Now, Consider the edge multiset on $\mathbb{F}_2^{n+1}$ defined by

$$\widehat{E} = \{(u0, v1), (v1, u0) : (u, v) \in E\}.$$

Obviously $(\mathbb{F}_2^{n+1}, \widehat{E}) \cong (\mathbb{F}_2^n, E) \otimes \mathbb{F}_2$. We map injectively each edge $(ua, vb) \in F$ to either the edge $(ua, vb) \in \widehat{E}$, if $a \neq b$; or to the path of length 2 $(ua, v(1-a)), (v(1-a), va)$, in case $a = b$. This injective multi mapping covers at least a quarter of the edges in $\widehat{E}$, and therefore by applying the 2^{p-1} -approximate triangle inequality,

$$\mathbb{E}_F \|f(x) - f(y)\|^p \leq 4 \cdot 2^p \cdot \mathbb{E}_{\widehat{E}} \|f(x) - f(y)\|^p.$$

Hence,

$$\gamma((\mathbb{F}_2^{n+1}, \widehat{E})/D^\perp, \|\cdot\|^p) \leq 4 \cdot 2^p \cdot \gamma((\mathbb{F}_2^n, E)/D^\perp, \|\cdot\|^p) = O(1).$$

But from our construction $(\mathbb{F}_2^{n+1}, \widehat{E})/D^\perp = ((\mathbb{F}_2^n, E)/C^\perp) \otimes \mathbb{F}_2$, so by applying Lemma 2.11,

$$\bar{\gamma}((\mathbb{F}_2^n, \widehat{E})/C^\perp, \|\cdot\|^p) \leq 2 \cdot \gamma((\mathbb{F}_2^n/C^\perp, \widehat{E}) \otimes \mathbb{F}_2^n, \|\cdot\|^p) = O(1).$$

Note also that the degree of $(\mathbb{F}_2^n/C^\perp, \widehat{E})$ is at most $4d_0$.

We are left to argue that the graph $(\mathbb{F}_2^n, E)$ (and hence $(\mathbb{F}_2^n, E)/C^\perp$) is a Cayley graph of the group $\mathbb{F}_2^n$. Let $F_0 \subset \mathbb{F}_2^{n+1}$ be the set of generators of the Cayley graph $(\mathbb{F}_2^{n+1}, F)$. Define

$$E_0 = \{u \in \mathbb{F}_2^n : \text{for each } a \in \mathbb{F}_2, ua \in F_0\} \cup 2d_0 \times \{\text{id}\}.$$

It is easy to check that E_0 is the set of generators of $(\mathbb{F}_2^n, E)$

Regarding the parity of the resulting graph: The number of vertices is a non-trivial divisor of 2^{n+1} and hence necessarily a power of 2. To make the degree of the graph even, we can double each of its edges, according to Lemma 2.12. $\square$

4. GRAPH PRODUCTS (FIXING ZIGZAG)

Definition 4 (Rotation map [26]). Let $G = (V, E)$ be an (n, d) graph, and $V = [n]$. A rotation map of the edges is $\text{rot}_G : [n] \times [d] \rightarrow [n] \times [d]$, such that $\text{rot}_G \circ \text{rot}_G = \text{id}$, and letting $\text{rot}_l : [n] \times [d] \rightarrow [n]$, $\text{rot}_r : [n] \times [d] \rightarrow [d]$ such that $\text{rot} = (\text{rot}_l, \text{rot}_r)$, we have $E = \{(u, \text{rot}_l(u, i)) : u \in [n], i \in [d]\}$

Let $G = (V_G, E_G)$ be a (n_1, d_1) multi-graph, and $H = (V_H, E_H)$ and (n_2, d_2) graph. The the tensor product $G \otimes H$ is an $(n_1 n_2, d_1 d_2)$ graph whose vertex-set is $V_{G \otimes H} = V_G \times V_H$, and whose edge set is

$$E_{G \otimes H} = \{((u_1, u_2), (v_1, v_2)) : (u_1, v_1) \in E_G, (u_2, v_2) \in E_H\}.$$

Proposition 4.1. Let ρ be a symmetric kernel on X , then

$$\bar{\gamma}(G \otimes H, \rho) \leq \bar{\gamma}(G, \rho) \cdot \bar{\gamma}(H, \rho).$$

Proof. Fix $f, g : V_G \times V_H \rightarrow (X, \rho)$. Write $\bar{\gamma}_G = \bar{\gamma}(G, \rho)$, and $\bar{\gamma}_H = \bar{\gamma}(H, \rho)$. So,

$$\begin{aligned} \mathbb{E}_{\substack{x, y \in V_G \\ i, j \in V_H}} \rho(f(x, i), g(y, j)) &= \mathbb{E}_{i, j \in V_H} \mathbb{E}_{x, y \in V_G} \rho(f(x, i), g(y, j)) \\ &\leq \bar{\gamma}_G \mathbb{E}_{i, j \in V_H} \mathbb{E}_{(x, y) \in E_G} \rho(f(x, i), g(y, j)) \\ &= \bar{\gamma}_G \mathbb{E}_{(x, y) \in E_G} \mathbb{E}_{i, j \in V_H} \rho(f(x, i), g(y, j)) \\ &\leq \bar{\gamma}_G \bar{\gamma}_H \mathbb{E}_{\substack{(x, y) \in E_G \\ (i, j) \in E_H}} \rho(f(x, i), g(y, j)) \\ &= \bar{\gamma}_G \bar{\gamma}_H \mathbb{E}_{((x, i), (y, j)) \in E(G \otimes H)} \rho(f(x, i), g(y, j)). \end{aligned} \quad \square$$

Remark 2. In terms of eigenvalues $\lambda(G \otimes H) = \min\{\lambda(G), \lambda(H)\}$, which means that

$$\bar{\gamma}(G \otimes H, (\mathbb{R}, |\cdot|^2)) = \max\{\bar{\gamma}(G, (\mathbb{R}, |\cdot|^2)), \bar{\gamma}(H, (\mathbb{R}, |\cdot|^2))\}.$$

The tensor product is used in [26] to obtain a computationally efficient expander constructions. The worse bound on $\bar{\gamma}(G \otimes H, \rho)$ here does not change much their construction.

Definition 5 (Zigzag product [26]). Let $G = (V_G = [n], E_G)$ be a (n, d_1) graph, and $H = (V_H = [d_1], E_H)$ a (d_1, d_2) graph. Define the zig-zag product $G \mathbin{\text{\textcircled{Z}}} H$ to be a (nd_1, d_2^2) graph whose vertex set is $V_G \times V_H$. The edge set is

$$E(G \mathbin{\text{\textcircled{Z}}} H) = \{((u, i), (\text{rotl}_G(u, \text{rotl}_H(i, \ell)), \text{rotl}_H(\text{rotl}_H(i, \ell), \ell')))) : (u, i) \in V_G \times V_H, \ell, \ell' \in [d_2]\}.$$

«In this notation, it's not entirely evident that for each edge in $E(G \mathbin{\text{\textcircled{Z}}} H)$ the reverse edge is also included (which it is). Should we show it explicitly, or at least remark it? —Manor»

MM

Proposition 4.2. *Let ρ be a symmetric kernel on X , then*

$$\bar{\gamma}(G \mathbin{\text{\textcircled{Z}}} H, \rho) \leq \bar{\gamma}(G, \rho) \cdot \bar{\gamma}(H, \rho)^2.$$

Proof. For simplicity of the notation, we assume that $V_G = [n_1]$, and $V_H = [d_1]$.

We further assume that the rotation map $\text{rot}_G : [n_1] \times [d_1] \rightarrow [n_1] \times [d_1]$ is *consistent*, which by this we mean that for every $v \in [n_1]$, $\{\ell \in [d_1] : \exists u \in [n_1] \text{rotl}_G(u, \ell) = v\} = [d_1]$. This means that we can now define $\text{rotR}_G : [n_1] \times [d_1] \rightarrow [d_1]$ to satisfy $\text{rotl}_G(\text{rotl}_G(v, p), \text{rotR}(v, p)) = v$.

Note that any (n, d) undirected multi-graph $G = (V, E)$ has a consistent labeling. To see this standard fact, consider the double cover of G , $G' = G \otimes \tau$. G' is d -regular bipartite graph. For every $A \subseteq V$, we claim that the neighborhood of $A \times \{0\}$, $N(A \times \{0\}) = \{(v, 1) : (u, v) \in E\}$, satisfies $|N(A \times \{0\})| \geq |A \times \{0\}|$. Indeed, assume that $|A \times \{0\}| > |N(A \times \{0\})|$. The number of edges going out from $A \times \{0\}$ is $d \cdot |A \times \{0\}|$. Hence the average degree is at least $d \cdot |A \times \{0\}| / |N(A \times \{0\})| > d$, which contradicts G' being d -regular. By Hall's marriage theorem, G' contains a perfect matching. Label this perfect matching by " d ", remove it and continue by induction on the resulting $(n, d - 1)$ graph. The end result is a consistent labeling.

Fix $f, g : V_G \times V_H \rightarrow (X, \rho)$. Write $\bar{\gamma}_G = \bar{\gamma}(G, \rho)$, and $\bar{\gamma}_H = \bar{\gamma}(H, \rho)$. So,

$$\begin{aligned}
& \mathbb{E}_{\substack{u, v \in [n_1] \\ k, \ell \in [d_1]}} \rho(f(u, k), g(v, \ell)) \\
&= \mathbb{E}_{k, \ell \in [d_1]} \mathbb{E}_{u, v \in [n_1]} \rho(f(u, k), g(v, \ell)) \\
&\leq \bar{\gamma}_G \mathbb{E}_{k, \ell \in [d_1]} \mathbb{E}_{\substack{u \in [n_1] \\ p \in [d_1]}} \rho(f(u, k), g(\text{rotl}_G(u, p), \ell)) \\
&= \bar{\gamma}_G \mathbb{E}_{\substack{u \in [n_1] \\ \ell \in [d_1]}} \mathbb{E}_{\substack{k \in [d_1] \\ p \in [d_1]}} \rho(f(u, k), g(\text{rotl}_G(u, p), \ell)) \\
&\leq \bar{\gamma}_G \bar{\gamma}_H \mathbb{E}_{\substack{u \in [n_1] \\ \ell \in [d_1]}} \mathbb{E}_{\substack{p \in [d_1] \\ j \in [d_2]}} \rho(f(u, \text{rotl}_H(p, j)), g(\text{rotl}_G(u, p), \ell)) \\
&= \bar{\gamma}_G \bar{\gamma}_H \mathbb{E}_{\substack{j \in [d_2] \\ \ell \in [d_1]}} \mathbb{E}_{\substack{p \in [d_1] \\ u \in [n_1]}} \rho(f(u, \text{rotl}_H(p, j)), g(\text{rotl}_G(u, p), \ell)) \\
&= \bar{\gamma}_G \bar{\gamma}_H \mathbb{E}_{\substack{j \in [d_2] \\ \ell \in [d_1]}} \mathbb{E}_{\substack{p \in [d_1] \\ v \in [n_1]}} \rho(f(\text{rotl}_G(v, \text{rotR}_G(v, p)), \text{rot}_H(p, j)), g(v, \ell)) \\
&= \bar{\gamma}_G \bar{\gamma}_H \mathbb{E}_{\substack{j \in [d_2] \\ v \in [n_1]}} \mathbb{E}_{\substack{p \in [d_1] \\ \ell \in [d_1]}} \rho(f(\text{rotl}_G(v, \text{rotR}_G(v, p)), \text{rotl}_H(p, j)), g(v, \ell)) \\
&\leq \bar{\gamma}_G \bar{\gamma}_H^2 \mathbb{E}_{\substack{j \in [d_2] \\ v \in [n_1]}} \mathbb{E}_{\substack{p \in [d_1] \\ i \in [d_2]}} \rho(f(\text{rotl}_G(v, \text{rotR}_G(v, p)), \text{rotl}_H(p, j)), g(v, \text{rotl}_H(p, i))) \\
&= \bar{\gamma}_G \bar{\gamma}_H^2 \mathbb{E}_{\substack{j \in [d_2] \\ u \in [n_1]}} \mathbb{E}_{\substack{p \in [d_1] \\ i \in [d_2]}} \rho(f(u, \text{rotl}_H(p, j)), g(\text{rotl}_G(u, p), \text{rotl}_H(p, i))) \\
&= \bar{\gamma}_G \bar{\gamma}_H^2 \mathbb{E}_{\substack{j \in [d_2] \\ u \in [n_1]}} \mathbb{E}_{\substack{k \in [d_1] \\ i \in [d_2]}} \rho(f(u, k), g(\text{rotl}_G(u, \text{rotl}_H(k, j)), \text{rotl}_H(\text{rot}_H(k, j), i))) \tag{12} \\
&= \bar{\gamma}_G \bar{\gamma}_H^2 \mathbb{E}_{((u, k), (v, \ell)) \in E(G \otimes H)} \rho(f(u, k), g(v, \ell)). \quad \square
\end{aligned}$$

Remark 3. The bound proved in Proposition 4.2 matches the best known bound [28, 25] on the spectral gap of the zigzag product (i.e., when ρ is square of line metric).

Definition 6 (Replacement product). Let $G = (V_G = [n], E_G)$ be a (n_1, d_1) graph, and $H = (V_H = [d_1], E_H)$ a (d_1, d_2) graph. Define the replacement product $G \oplus H$ to be a $(nd_1, d_2 + 1)$ graph whose vertex set is $V_G \times V_H$. The edge set is

$$E(G \oplus H) = \{((u, i), (u, j)) : u \in V_G, (i, j) \in E_H\} \cup \{((u, i), (\text{rot}_G(u, i), i)) : u \in V_G, i \in V_H\}.$$

The balanced replacement product is similar but with d_2 copies of the later edges.

Definition 7 (Balanced replacement product [26]). Let $G = (V_G = [n], E_G)$ be a (n_1, d_1) graph, and $H = (V_H = [d_1], E_H)$ a (d_1, d_2) graph. Define the balanced replacement product $G \circledast H$ to be a (nd_1, d_2) graph whose vertex set is $V_G \times V_H$. The edge set is

$$E(G \circledast H) = \{((u, i), (u, j)) : u \in V_G, (i, j) \in E_H\} \cup \{d_2 \times ((u, i), (\text{rot}_G(u, i), i)) : u \in V_G, i \in V_H\}.$$

The analysis replacement and balanced replacement products requires that ρ satisfies the approximate triangle inequality. The proof essentially reduces to the zigzag case

Proposition 4.3. *Let ρ be a c -approximate metric on X . Then*

$$\begin{aligned}
\bar{\gamma}(G \circledast H, \rho) &\leq 4c^2 \bar{\gamma}(G, \rho) \cdot \bar{\gamma}(H, \rho)^2, \\
\bar{\gamma}(G \oplus H, \rho) &\leq \max\{4, d_2\} c^2 \bar{\gamma}(G, \rho) \cdot \bar{\gamma}(H, \rho)^2,
\end{aligned}$$

where H has degree d_2 .

Proof. For simplicity of the notation, we assume that $V_G = [n_1]$, and $V_H = [d_1]$.

Fix $f, g : V_G \times V_H \rightarrow (X, \rho)$. Write $\bar{\gamma}_G = \bar{\gamma}(G, \rho)$, and $\bar{\gamma}_H = \bar{\gamma}(H, \rho)$.

Consider a zigzag edge $((u, i), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(\text{rot}_H(i, \ell), \ell')))$. It can be written a path of length three in the replacement/balanced replacement product:

$$\begin{aligned} ((u, i), (u, \text{rot}_H(i, \ell))), & \quad ((u, \text{rot}_H(i, \ell)), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(i, \ell))), \\ & \quad ((\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(i, \ell)), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(\text{rot}_H(i, \ell), \ell')))). \end{aligned} \quad (13)$$

and therefore $\ll$ HERE WE LOSE A BIT BECAUSE OF USING ABSTRACT c -approximate triangle inequality $\gg$ MM instead of power of metric. We could have used weighted Holder inequality —Manor $\gg$

$$\begin{aligned} & \rho((u, i), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(\text{rot}_H(i, \ell), \ell'))) \\ & \leq c^2 \left(\rho((u, i), (u, \text{rot}_H(i, \ell))) + \rho((u, \text{rot}_H(i, \ell)), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(i, \ell))) \right. \\ & \quad \left. + \rho((\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(i, \ell)), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(\text{rot}_H(i, \ell), \ell'))) \right). \end{aligned}$$

In the balanced replacement product the first and third edge in (13) are used by $2d_2$ zigzag edges each, whereas the middle one uses d_2 times. Hence they are balanced and we can continue from (12):

$$\begin{aligned} \frac{(12)}{\bar{\gamma}_G \bar{\gamma}_H^2} &= \frac{1}{n_1 d_1 d_2^2} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \sum_{\substack{\ell \in [d_2] \\ \ell' \in [d_2]}} \rho((u, i), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(\text{rot}_H(i, \ell), \ell'))) \\ &\leq \frac{c^2}{n_1 d_1 d_2^2} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \sum_{\substack{\ell \in [d_2] \\ \ell' \in [d_2]}} \rho((u, i), (u, \text{rot}_H(i, \ell))) \\ &\quad + \frac{c^2}{n_1 d_1 d_2^2} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \sum_{\substack{\ell \in [d_2] \\ \ell' \in [d_2]}} \rho((u, \text{rot}_H(i, \ell)), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(i, \ell))) \\ &\quad + \frac{c^2}{n_1 d_1 d_2^2} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \sum_{\substack{\ell \in [d_2] \\ \ell' \in [d_2]}} \rho((\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(i, \ell)), (\text{rot}_G(u, \text{rot}_H(i, \ell)), \text{rot}_H(\text{rot}_H(i, \ell), \ell'))) \\ &= \frac{c^2}{n_1 d_1 d_2} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \sum_{\ell \in [d_2]} \rho((u, i), (u, \text{rot}_H(i, \ell))) \\ &\quad + \frac{c^2}{n_1 d_1} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \rho((u, i), (\text{rot}_G(u, i), i)) \\ &\quad + \frac{c^2}{n_1 d_1 d_2} \sum_{\substack{i \in [d_1] \\ u \in [n_1]}} \sum_{\ell' \in [d_2]} \rho((u, i), (u, \text{rot}_H(i, \ell'))) \\ &\leq 4c^2 \mathbb{E}_{(a,b) \in E_{G \oplus H}} \rho(a, b). \end{aligned} \quad (14)$$

$$\leq 4c^2 \mathbb{E}_{(a,b) \in E_{G \oplus H}} \rho(a, b). \quad (15)$$

The bound on the replacement product similar, but now in (14) every edge is weighted d_2 more than it appearance in the graph $G \oplus H$. So (15) is replaced with the upper bound

$$\max\{4, d_2\} c^2 \mathbb{E}_{(a,b) \in E_{G \oplus H}} \rho(a, b). \quad \square$$

Remark 4. The analysis used here is much simpler than the linear algebra analysis, and much more general. The balanced replacement was previously analyzed with the edge expansion parameter (which is approximately $\gamma(\cdot, (\mathbb{R}, |\cdot|))$, see Proposition 2.4 in [2].

Remark 5. We have abused the notation of the zigzag and replacement products, since the definition requires the rotation map, one can have different non-isomorphic graph from different rotation maps. However, the bounds on $\bar{\gamma}$, and the degrees are independent of the rotation maps, which is sufficient for our purposes.

5. GRAPH POWERS

The iterative construction of zigzag expanders of [26] requires an operation $\mathcal{O}$ that improves the “spectral gap”. In the context of “classic” spectral gap, i.e. when $\rho = \|\cdot\|_2^2$, the natural candidate is graph power. In this section we explore sufficient geometric conditions on ρ for which (variants of) graph power improves the spectral gap in terms of $\bar{\gamma}$. We prove it for two geometric conditions: When $\rho = \|\cdot\|^p$ and $\|\cdot\|$ is a p -convex norm, and when ρ is a square of CAT(0) metric. Unfortunately in both cases, we are unable to prove it to the vanilla graph power, and resort to proving to a soup-up variant of graph power. However this variant is sufficient to construct expanders.

In Section 5.2 we prove it for a common generalization of CAT(0) spaces and uniform convex Banach spaces, which we call p -barycentric spaces. In Section 5.1, we prove for the special case of uniform convex space. While the argument is essentially the same, the notation in this case is much simpler, and for the sake of clarity we present it first. In Section 5.3 we prove the actual decay of $\bar{\gamma}$ as a consequence of the previous two sections.

Our methods are inspired by the Markov cotype property of Ball [5], and this will be picked-up later in Section 7

5.1. Uniform convex spaces.

A normed space $(X, \|\cdot\|)$ is called p convex with constant K if for every $x, y \in X$,

$$2\|x\|^p + \frac{2}{K^p}\|y\|^p \leq \|x+y\|^p + \|x-y\|^p.$$

Lemma 5.1 (cf. [5, Lemma 3.1]). *Let X be p -convex space with constant $K > 0$, and U a random vector in $L_p(X)$. Then*

$$\|\mathbb{E}U\|^p + K^{-p}\mathbb{E}\|U - \mathbb{E}U\|^p \leq \mathbb{E}\|U\|^p.$$

«Ball proves it only to $q = 2$. to check generalization —Manor»

Pisier [22] used it to prove.

Lemma 5.2. *Let X be p -convex space with constant $K > 0$, and $M_0, \dots, M_m$ a martingale in $L_p(X)$. Then*

$$\sum_{t=0}^{m-1} \mathbb{E}\|M_{t+1} - M_t\|^p \leq K^p \mathbb{E}\|M_m - M_0\|^p.$$

Denote $\mathcal{A}_m(B) = \frac{1}{m}(\sum_{t=0}^{m-1} B^t)$.

Lemma 5.3. *Suppose $(X, \|\cdot\|)$ is p -convex Banach space with constant $K > 0$. Then for every $n \in \mathbb{N}$, $m \in \mathbb{N}$, symmetric stochastic $n \times n$ matrix A , and every $x_1, \dots, x_n \in X$,*

$$m \sum_{i,j} A_{ij} \|y_i - y_j\|^p + \sum_i \|x_i - y_i\|^p \leq (19K)^p \sum_{i,j} \mathcal{A}_m(A)_{ij} \|x_i - x_j\|^p, \quad (16)$$

where $y_i = \sum_j \mathcal{A}_m(A)_{ij} x_j$.

Proof. We make use of Pisier’s “Martingale cotype” equivalent definition of p -convexity as stated in Lemma 5.2. For every $\ell \in [n]$ we form a martingale $(M_t^\ell)_{t=0}^m$ by

$$\Pr \left(M_t^\ell = \sum_k (A^{m-t})_{kj} x_k \right) = (A^t)_{j\ell}.$$

This is clearly a martingale, and so by Lemma 5.2,

$$\sum_t \sum_i (A^t)_{\ell i} \sum_j A_{ij} \left\| \sum_k (A^{m-t+1})_{ik} x_k - \sum_k A_{jk}^{m-t} x_k \right\|^p \leq K^p \sum_j (A^m)_{\ell j} \|x_j - \sum_k (A^m)_{\ell k} x_k\|^p \quad (17)$$

Summing the left hand side of (17) over $\ell \in [n]$, we obtain

$$\begin{aligned}
& \sum_{t \in [m]} \sum_{i \in [n]} \sum_{\ell \in [n]} (A^t)_{\ell i} \sum_j A_{ij} \left\| \sum_k (A^{m-t+1})_{ik} x_k - \sum_k A_{jk}^{m-t} x_k \right\|^p \\
&= \sum_t \sum_i \sum_j A_{ij} \left\| \sum_k (A^{m-t+1})_{ik} x_k - \sum_k A_{jk}^{m-t} x_k \right\|^p \\
&\geq m \sum_{ij} A_{ij} \left\| \sum_k (A \mathcal{A}_m(A))_{ik} x_k - \sum_k \mathcal{A}_m(A)_{jk} x_k \right\|^p \\
&= m \sum_{ij} A_{ij} \left\| \sum_k (\mathcal{A}_m(A) - \frac{I}{m} + \frac{A^m}{m})_{ik} x_k - \sum_k (\mathcal{A}_m(A))_{jk} x_k \right\|^p \\
&\geq 3^{-p} m \sum_{ij} A_{ij} \left\| \sum_k (\mathcal{A}_m(A))_{ik} x_k - \sum_k (\mathcal{A}_m(A))_{jk} x_k \right\|^p - m^{1-p} \sum_i \left\| x_i - \sum_k (A^m)_{ik} x_k \right\|^p \\
&\geq 3^{-p} m \sum_{ij} A_{ij} \left\| \sum_k (\mathcal{A}_m(A))_{ik} x_k - \sum_k (\mathcal{A}_m(A))_{jk} x_k \right\|^p - m^{1-p} \sum_{ij} (A^m)_{ij} \|x_i - x_j\|^p. \tag{18}
\end{aligned}$$

Next we sum over ℓ the right hand side of (17), and obtain

$$\begin{aligned}
& \sum_{ij} (A^m)_{ij} \|x_j - \sum_k (A^m)_{ik} x_k\|^p \\
&\leq \sum_{ijk} (A^m)_{ij} (A^m)_{ik} \|x_j - x_k\|^p \\
&\leq 2^{p-1} \sum_{ijk} (A^m)_{ij} (A^m)_{ik} (\|x_j - x_i\|^p + \|x_i - x_k\|^p) \\
&= 2^p \sum_{ij} A_{ij}^m \|x_i - x_j\|^p \tag{19}
\end{aligned}$$

Combining (17) with (18) and (19), we obtain

$$\begin{aligned}
& m \sum_{ij} A_{ij} \left\| \sum_k (\mathcal{A}_m(A))_{ik} x_k - \sum_k (\mathcal{A}_m(A))_{jk} x_k \right\|^p \\
&\leq ((6K)^p + 3^p) \sum_{ij} A_{ij}^m \|x_i - x_j\|^p \\
&= ((6K)^p + 3^p) \sum_{ij} \frac{1}{m} \left(\sum_{t=0}^{m-1} A^t A^{m-t} \right)_{ij} \|x_i - x_j\|^p \\
&\leq 2^{p-1} ((6K)^p + 3^p) \sum_{ijk} \frac{1}{m} \left(\sum_{t=0}^{m-1} A_{ik}^t A_{jk}^{m-t} \right) (\|x_k - x_i\|^p + \|x_j - x_k\|^p) \\
&= 2^p ((6K)^p + 3^p) \sum_{ij} \mathcal{A}(A)_{ij} \|x_i - x_j\|^p. \tag{20}
\end{aligned}$$

Also,

$$\sum_i \|x_i - \mathcal{A}_m(A)_{ij} x_j\|^p \leq \sum_{ij} \mathcal{A}_m(A)_{ij} \|x_i - x_j\|^p.$$

Assigning $y_i = \sum_j \mathcal{A}_m(A)_{ij} x_j$, and summing the last inequality with (20) we conclude that

$$m \sum_{ij} a_{ij} \|y_i - y_j\|^p + \sum_i \|x_i - y_i\|^p \leq (1 + 2^p ((6K)^p + 3^p)) \sum_{ij} \mathcal{A}_m(A)_{ij} \|x_i - x_j\|^p. \quad \square$$

5.2. **p -Barycentric Spaces.** «should we call those spaces p -barycentric with constant K ? —Manor»

Definition 8. A metric space (X, ρ) is called having p -barycentric with constant $K \geq 1$ if for every random variable «what technical limitation this rv should have? —Manor» $U : \Omega \rightarrow X$, there exists a point $\text{cm}(U)$ such that

- For every probability space (Ω, μ) and any two random variables $U, V : \Omega \rightarrow X$ on this probability space,

$$\rho(\text{cm}(U), \text{cm}(V)) \leq \mathbb{E}_\mu \rho(U, V). \quad (21)$$

- For every random variable $U : (\Omega, \mu) \rightarrow X$, and for every $x \in X$,

$$\rho(\text{cm}(U), x)^p + K^{-p} \mathbb{E} \rho(U, \text{cm}(U))^p \leq \mathbb{E} \rho(U, x)^p. \quad (22)$$

«Do we want this common generalization of p -convex and $\text{CAT}(0)$? It will sure save the need to develop Markov cotype and sarkozy cotype twice. However, it obfuscate the argument in p -convex space. Also having define it, it make sense to study those objects. I'm sure they are uniquely geodesics for example (for every $a, b \in X$, and $t \in [0, 1]$ let $[a, b]_t$ be the center of mass of the random variable that has probability t being in a and probability $1 - t$ being in b . Then $[a, b]_{t \in [0, 1]}$ is probably the geodesic). This may be the starting point for the common generalization of p -convex and $\text{CAT}(0)$ that Assaf claimed. —Manor»

Remark 6. Property (22) implies that the center of mass is unique for a given p (for any constant K).

The following is well known

Proposition 5.4. *CAT(0) is having 2-barycentric with constant 1.*

Proof. Condition (22) is well known, see e.g., [15, Lemma 4.1] «TO ADD A CITATION FORM A BOOK —Manor». Condition (21) is also true, see [15, Lemma 4.2]. $\square$

The following is also well known

Proposition 5.5. *A p -convex Banach space with constant K is p -barycentric with constant K .*

Proof. Define $\text{cm}(U) = \mathbb{E} U$. Condition (22) is proved in [5, Lemma 3.1] «It is proved there only for 2-convex. Check that it generalizes —Manor», and (21) follows from the convexity of the norm. $\square$

Definition 9 (Martingales in barycentric spaces). Let (X, ρ) be a p -barycentric space. A martingale on T with filterization $\mathcal{F} = (F_i)_i$ as a series of random variables in X , $(M_i)_i$, M_i is defined by the σ -algebra F_i , F_i refines F_{i-1} , F_0 is a singleton, and $\text{cm}[M_i | F_{i-1}] = M_{i-1}$.

Martingales in $\text{CAT}(0)$ spaces are studied in [30]. Note that, unlike martingales in uniformly convex spaces, in general it is not assured that $\text{cm}[M_i | F_j] = M_j$, when $j < i - 1$. Nevertheless, this definition is sufficient for proving a martingale cotype p inequality for p -barycentric spaces, abstracting the martingale cotype p inequality for p -convex spaces of Pisier [22].

Lemma 5.6. *If $(M_t)_{t=0}^m$ is a martingale on p -barycentric space (X, ρ) with constant $K \geq 1$. Then,*

$$\sum_{t=0}^{m-1} \mathbb{E} \rho(M_{t+1}, M_t)^p \leq K^p \mathbb{E} \rho(M_m, M_0)^p.$$

Proof. Apply (22) with $U = [M_k | M_{k-1}]$, and $x = M_0$, we obtain

$$\rho(M_{k-1}, M_0)^p + K^{-p} \mathbb{E}[\rho(M_k, M_{k-1})^p | M_{k-1}] \leq \mathbb{E}[\rho(M_k, M_0)^p | M_{k-1}].$$

Taking the expectation with respect to M_{k-1} we obtain

$$\mathbb{E}[\rho(M_k, M_{k-1})^p] \leq K^p (\mathbb{E}[\rho(M_k, M_0)^p] - \mathbb{E}[\rho(M_{k-1}, M_0)^p]).$$

By summing the last inequality over $k \in \{1, \dots, m\}$, we conclude the claim. $\square$

Proposition 5.7. *Suppose (X, ρ) is a p -barycentric space with constant $K \geq 1$. Then for every $n \in \mathbb{N}$, $m \in \mathbb{N}$, symmetric stochastic $n \times n$ matrix A , and every $x_1, \dots, x_n \in X$, there exist $y_1, \dots, y_n \in X$ such that*

$$m \sum_{ij} A_{ij} \rho(y_i, y_j)^p + \sum_i \rho(x_i, y_i)^p \leq (10K)^p \sum_{i,j} A_m(A)_{ij} \rho(x_i, x_j)^p. \quad (23)$$

Proof. To simplify the notation, we use the notation $\text{cm}(\frac{1}{m} \sum_{i=1}^m x_i)$ which means the center of mass of the r.v. that get with probability $1/m$ the point x_i .

For a fixed $\ell \in [n]$, $m \in \mathbb{N}$, and $x_1, \dots, x_n \in X$, denote $(M_t^{\ell, m})_{t=0}^m$ the X -valued pseudo martingale, defined on the filterization $(\mathcal{F}_t)_{t=0}^m = (i_1, \dots, i_t) \in [n]^t$ such that $\Pr[(i_1, \dots, i_t)] = \prod_{j=1}^t A_{i_{j-1}, i_j}$, where $i_0 = \ell$. The values of the pseudo martingale $M_t^{\ell, m}$ are defined by reverse induction as follows

$$M_m^{\ell, m}(i_1, \dots, i_m) = x_{i_m},$$

and assuming $M_{t+1}^{\ell, m}$ was defined

$$M_t^{\ell, m}(i_1, \dots, i_t) := \text{cm} \left(\sum_j A_{i_t, j} M_{t+1}^{\ell, m}(i_1, \dots, i_t, j) \right),$$

where the sum of pseudo-martingale values is a convex sum and is interpreted as the the center of mass according to those weights. Note that for $j \leq t \leq m$, $M_t^{\ell, m}(i_1, \dots, i_t) = M_{t-j}^{i_j, m-j}(i_{j+1}, \dots, i_t)$. In particular, $M_t^{\ell, m}(i_1, \dots, i_t) = M_0^{i_t, m-t}$.

This is clearly a pseudo-martingale, and so, fixing $\ell \in [n]$, by Lemma 5.6,

$$\sum_{t \in [m]} \sum_{i \in [n]} (A^t)_{\ell i} \sum_{j \in [n]} A_{ij} \rho \left(M_0^{i, m-t-1}, M_0^{j, m-t} \right)^p \leq K^p \sum_{j \in [n]} (A^m)_{\ell j} \rho(x_j, M_0^{\ell, m})^p. \quad (24)$$

Summing the left hand side of (24) over $\ell \in [n]$, we obtain

$$\begin{aligned} & \sum_{t \in [m]} \sum_{i \in [n]} \sum_{\ell \in [n]} (A^t)_{\ell i} \sum_j A_{ij} \rho \left(M_0^{i, m-t+1}, M_0^{j, m-t} \right)^p \\ &= \sum_{t \in [m]} \sum_{i, j \in [n]} A_{ij} \rho \left(M_0^{i, m-t+1}, M_0^{j, m-t} \right)^p \\ &= m \sum_{i, j \in [n]} A_{ij} \sum_{t \in [m]} \frac{1}{m} \cdot \rho(M_0^{i, t}, M_0^{j, t-1})^p \\ &\geq m \sum_{i, j \in [n]} A_{ij} \left(\sum_{t \in [m]} \frac{1}{m} \cdot \rho(M_0^{i, t}, M_0^{j, t-1}) \right)^p \end{aligned} \quad (25)$$

$$\geq m \sum_{i, j \in [n]} A_{ij} \rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{i, t} \right), \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j, t+1} \right) \right)^p \quad (26)$$

$$\geq 2^{-p} m \sum_{i, j \in [n]} A_{ij} \rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{i, t} \right), \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j, t} \right) \right)^p - \sum_{j \in [n]} \frac{\rho(M_0^{j, m}, x_j)^p}{m^{p-1}}. \quad (27)$$

Inequality (25) follows from the convexity of power $p \geq 1$, (26) follows from (21) and (27) is followed from the following inequality:

$$\begin{aligned}
& \rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{i,t}, \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t} \right) \right)^p \right. \\
& \leq 2^p \rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{i,t}, \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t+1} \right) \right)^p + 2^p \rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t}, \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t+1} \right) \right)^p \right),
\end{aligned}$$

and

$$\rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t}, \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t+1} \right) \right)^p \leq \left(\frac{\rho(M_0^{j,m}, M_0^{j,0})}{m} \right)^p \quad \text{By (21)}$$

In general, by iteratively applying (21), we have

$$\begin{aligned}
\rho(z, M_0^{\ell,m}) & \leq \sum_{i_1 \in [n]} A_{\ell i_1} \rho(z, M_1^{i_1,m}) \\
& \leq \sum_{i_1, i_2 \in [n]} A_{\ell i_1} A_{i_1 i_2} \rho(z, M_2^{i_2,m}) \\
& = \sum_{i_2 \in [n]} (A^2)_{\ell i_2} \rho(z, M_2^{i_2,m}) \\
& \vdots \\
& \leq \sum_{i \in [n]} (A^m)_{\ell i} \rho(z, x_i).
\end{aligned}$$

So summing over ℓ in the right hand side of (24), we obtain

$$\begin{aligned}
& \sum_{\ell j} (A^m)_{\ell j} \rho(x_j, M_0^{\ell,m})^p \\
& \leq \sum_{\ell j k} (A^m)_{\ell j} (A^m)_{\ell k} \rho(x_j, x_k)^p \tag{28}
\end{aligned}$$

$$\begin{aligned}
& = \sum_{j k} (A^{2m})_{j k} \rho(x_j, x_k)^p \\
& \leq 2^{p-1} \sum_{\ell j k} (A^m)_{\ell j} (A^m)_{\ell k} (\rho(x_j, x_\ell)^p + \rho(x_i, x_k)^p) \\
& = 2^p \sum_{i j} A_{ij}^m \rho(x_i, x_j)^p. \tag{29}
\end{aligned}$$

So combining (24) with (27) and (29) we obtain

$$\begin{aligned}
& m \sum_{i,j \in [n]} A_{ij} \rho \left(\text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{i,t} \right), \text{cm} \left(\sum_{t \in [m]} \frac{1}{m} \cdot M_0^{j,t} \right) \right)^p \\
& \leq 4^p K^p \sum_{ij} A_{ij}^m \rho(x_i, x_j)^p + \frac{2^p}{m^{p-1}} \cdot \sum_{j \in [n]} \rho(M_0^{j,m}, x_j)^p \\
& \leq ((4K)^p + 2^p) \sum_{ij} A_{ij}^m \rho(x_i, x_j)^p \\
& = ((4K)^p + 2^p) \sum_{ij} \frac{1}{m} \left(\sum_{t=0}^{m-1} A^t A^{m-t} \right)_{ij} \rho(x_i, x_j)^p \\
& \leq 2^{p-1} \cdot ((4K)^p + 2^p) \sum_{ijk} \frac{1}{m} \left(\sum_{t=0}^{m-1} A_{ik}^t A_{jk}^{m-t} \right) (\rho(x_k, x_i)^p + \rho(x_j, x_k)^p) \\
& = ((8K)^p + 2^p) \sum_{ij} \mathcal{A}_m(A)_{ij} \|x_i - x_j\|^p. \tag{30}
\end{aligned}$$

Again, by iteratively applying Proposition (21) we also have

$$\sum_i \rho(x_i, \text{cm}(\frac{1}{m} \sum_{t \in [m]} M_0^{i,t}))^p \leq \sum_{ij} \mathcal{A}_m(A)_{ij} \rho(x_i, x_j)^p.$$

Assigning $y_i = \text{cm}(\frac{1}{m} \sum_{t \in [m]} M_0^{i,t})$, and summing the last inequality with (30) we conclude that

$$m \sum_{i,j \in [n]} A_{ij} \rho(y_i, y_j)^p + \sum_i \rho(x_i, y_i)^p \leq ((8K)^p + 2^p + 1) \sum_{ij} \mathcal{A}_m(A)_{ij} \rho(x_i, x_j)^p. \quad \square$$

5.3. The decay of $\bar{\gamma}$. Denote by $\mathcal{A}_m(G)$, the natural graph whose the transition matrix of the random walk on G is $\mathcal{A}_m(A)$, where A is the transition matrix of the random walk on G . A formal definition

Definition 10. Let G be an undirected multi graph denote $E(G)$ the edges (multi)set of G , and by $\ell \times E(G)$ the multiset obtained by taking ℓ copies for each edge in $E(G)$. Then $\mathcal{A}_m(G)$ is a graph whose vertex set is the same as G , and whose edge multiset is

$$d^m \times E(G) \cup d^{m-1} \times E(G) \cup d^{m-2} \times E(G^2) \cup \dots \cup d^{m-1} \times E(G^{m-1}),$$

where the union is in the multiset sense.

Proposition 5.8. Let (X, ρ) be a c -approximate metric space satisfying (23). More precisely, it satisfies: For every $n \in \mathbb{N}$, $m \in \mathbb{N}$, symmetric stochastic $n \times n$ matrix A , and every $x_1, \dots, x_n \in X$, there exist $y_1, \dots, y_n$ such that

$$m \sum_i A_{ij} \rho(y_i, y_j) + \sum_i \rho(x_i, y_i) \leq K \sum_{i,j} \mathcal{A}_m(A)_{ij} \rho(x_i, x_j). \tag{31}$$

Then for every regular multigraph graph $G = (V, E)$,

$$\gamma(\mathcal{A}_m(G), \rho) \leq 2c^2 K \cdot \max\{1, \gamma(G, \rho)/m\}.$$

Proof. We should prove that for every non-constant $x : V \rightarrow X$, there exists $y : V \rightarrow X$ such that

$$\frac{\mathbb{E}_{V \times V} \rho(x_i, x_j)}{\mathbb{E}_{E(\mathcal{A}_m(G))} \rho(x_i, x_j)} \leq \frac{\mathbb{E}_{V \times V} \rho(y_i, y_j)}{\mathbb{E}_{E(G)} \rho(y_i, y_j)} \cdot \frac{4c^2 K}{m}. \tag{32}$$

We assume that there exists non-constant $x : V \rightarrow X$ such that

$$\frac{\mathbb{E}_{V \times V} \rho(x_i, x_j)}{\mathbb{E}_{E(\mathcal{A}_m(G))} \rho(x_i, x_j)} \geq 4c^2 K. \tag{33}$$

Hence we may restrict the universal quantifier on x in (32) to those satisfying (33).

By applying the c -approximate triangle inequality,

$$\rho(x_i, x_j) \leq c^2 \rho(x_i, y_i) + c^2 \rho(x_j, y_j)^p + c^2(y_i, y_j).$$

Averaging over $V \times V$, we obtain

$$\begin{aligned} \mathbb{E}_{V \times V} \rho(y_i, y_j) &\geq \frac{1}{c^2} \mathbb{E}_{V \times V} \rho(x_i, x_j) - 2 \mathbb{E}_V \rho(x_i, y_i) \\ &= \frac{1}{2 \cdot c^2} \mathbb{E}_{V \times V} \rho(x_i, x_j) + \frac{1}{2 \cdot c^2} \mathbb{E}_{V \times V} \rho(x_i, x_j) - 2 \mathbb{E}_V \rho(x_i, y_i) \\ &\geq \frac{1}{2 \cdot c^2} \mathbb{E}_{V \times V} \rho(x_i, x_j) + 2K_{E(\mathcal{A}_m(G))} \mathbb{E}_V \rho(x_i, x_j) - 2 \mathbb{E}_V \rho(x_i, y_i) && \text{Applying (33)} \\ &\geq \frac{1}{2 \cdot c^2} \mathbb{E}_{V \times V} \rho(x_i, x_j) && \text{Applying (31).} \end{aligned}$$

In particular it implies that y is non-constant. Now beginning with (31), and applying the inequality we have just deduced, we conclude

$$\frac{\mathbb{E}_{V \times V} \rho(x_i, x_j)}{\mathbb{E}_{E(\mathcal{A}_m(G))} \rho(x_i, x_j)} \leq \frac{K}{m} \cdot \frac{\mathbb{E}_{V \times V} \rho(x_i, x_j)}{\mathbb{E}_{E(G)} \rho(y_i, y_j)} \leq \frac{2c^2 K}{m} \cdot \frac{\mathbb{E}_{V \times V} \rho(y_i, y_j)}{\mathbb{E}_{E(G)} \rho(y_i, y_j)}. \quad \square$$

We actually need the decay of $\bar{\gamma}$. We can repeat the arguments in the section with double cover. However this would be very lengthy. Instead we choose the slightly less efficient (in terms of the multiplicative factors) but more civilized approach of reducing the decay $\bar{\gamma}$ to the decay of γ .

Proposition 5.9. *Assume that ρ is c -approximate metric on X for which for every regular multigraph graph $G = (V, E)$, $\gamma(\mathcal{A}_m(G), \rho) \leq K \cdot \max\{1, \gamma(G, \rho)/m\}$.*

Then for every regular multigraph graph $G = (V, E)$,

$$\bar{\gamma}(\mathcal{A}_m(G), \rho) \leq 48c^2 K \cdot \max\{1, \bar{\gamma}(G, \rho)/m\}.$$

Proof. We use the correspondence between $\bar{\gamma}(G)$ and $\gamma(G \otimes \tau)$, the double cover of G , as is shown in Lemma 2.11. We observe that for odd t , $(G \otimes \tau)^t = G^t \otimes \tau$. What complicates matters is that for even t , $(G \otimes \tau)^t$ is two disjoint copies of G^t . However, we can approximate even t as a sum of two almost unique odd numbers by defining for even t , $o_1(t) \leq o_2(t)$ the unique pair of odd numbers such that $o_2(t) - o_1(t) \leq 2$, and $o_1(t) + o_2(t) = t$. Note that the multiset

$$\{o_1(t), o_2(t) : t \text{ is even, and } t \leq m\}$$

contains odd numbers in the range $\{1, \dots, \lfloor m/2 \rfloor + 1\}$, and at most four items for each value. We therefore can bound for every $h : V \times \{0, 1\} \rightarrow X$,

$$\begin{aligned}
& \mathbb{E}_{\substack{(x,y) \in \\ E(\mathcal{A}_m(G \otimes \tau))}} \rho(h(x), h(y)) \\
&= \frac{1}{m} \sum_{t=0}^{m-1} \mathbb{E}_{\substack{(x,y) \in \\ E((G \otimes \tau)^t)}} \rho(h(x), h(y)) \\
&\leq \frac{1}{m} \sum_{\substack{t \in [m] \\ t \text{ is odd}}} \mathbb{E}_{E(G^t \otimes \tau)} \rho(h(x), h(y)) \\
&\quad + \frac{1}{m} \sum_{\substack{t \in [m] \\ t \text{ is even}}} c \left(\mathbb{E}_{E(G^{o_1(t)} \otimes \tau)} \rho(h(x), h(y)) + \mathbb{E}_{E(G^{o_2(t)} \otimes \tau)} \rho(h(x), h(y)) \right) \\
&\leq \frac{4c}{m} \sum_{\substack{t \in [m] \\ t \text{ is odd}}} \mathbb{E}_{E(G^t \otimes \tau)} \rho(h(x), h(y)) \\
&\leq \frac{12c}{m} \sum_{t=0}^{m-1} \mathbb{E}_{E(G^t \otimes \tau)} \rho(h(x), h(y)) \\
&= 12c \mathbb{E}_{\substack{(x,y) \in \\ E(\mathcal{A}_m(G) \otimes \tau)}} \rho(h(x), h(y))
\end{aligned}$$

Hence

$$\begin{aligned}
\mathbb{E}_{\substack{x,y \in \\ V \times \{0,1\}}} \rho(h(x), h(y)) &\leq \gamma(\mathcal{A}_m(G \otimes \tau)) \cdot \mathbb{E}_{\substack{(x,y) \in \\ E(\mathcal{A}_m(G \otimes \tau))}} \rho(h(x), h(y)) \\
&\leq 12c\gamma(\mathcal{A}_m(G \otimes \tau)) \cdot \mathbb{E}_{\substack{(x,y) \in \\ E(\mathcal{A}_m(G) \otimes \tau)}} \rho(h(x), h(y)).
\end{aligned}$$

We have just proved that

$$\gamma(\mathcal{A}_m(G) \otimes \tau) \leq 12c\gamma(\mathcal{A}_m(G \otimes \tau)). \quad (34)$$

We conclude:

$$\begin{aligned}
\bar{\gamma}(\mathcal{A}_m(G), \rho) &\leq 2\gamma(\mathcal{A}_m(G) \otimes \tau, \rho) && \text{By Lemma 2.11} \\
&\leq 24c\gamma(\mathcal{A}_m(G \otimes \tau), \rho) && \text{By (34)} \\
&\leq 24cK \max\{1, \gamma(G \otimes \tau, \rho)/m\} && \text{By the assumptions} \\
&\leq 48c^2K \max\{1, \bar{\gamma}(G, \rho)/m\} && \text{By Lemma 2.11}
\end{aligned}$$

□

Corollary 5.10. *Let (X, ρ) be a p -barycentric space with constant K . Then for every regular multigraph $G = (V, E)$,*

$$\bar{\gamma}(\mathcal{A}_m(G), \|\cdot\|^p) \leq 24(20K)^p \cdot \max\{1, \bar{\gamma}(G, \|\cdot\|^p)/m\}.$$

6. EXPANDERS

Lemma 6.1. *Let X be a p -convex Banach space. Then there exist $C = C(X) > 0$, $d_0 = d_0(X) \in \mathbb{N}$, and a family of (n_i, d_0^2) graphs $G_i = (V_i, E_i)$ where $n_i \nearrow \infty$, such that $\bar{\gamma}(G_i, \|\cdot\|_X^p) \leq C$.*

Proof. Since p -convex Banach space is also K -convex, by Corollary 3.10 there are constants $C_1 \geq 1$, $0.3 \geq \delta > 0$ depending only on X and p , and family of (N_i, D_i) graphs $\{B_i\}_{i \in \mathbb{N}}$ for unbounded N_i , such that $\bar{\gamma}(B_i, \|\cdot\|_X^p) \leq C_1$, and $D_i \leq \exp(\log^{1-\delta} N_i)$.

By Corollary 5.10, for every regular graph G , and $t \in \mathbb{N}$,

$$\bar{\gamma}(\mathcal{A}_t(G), \|\cdot\|_X^p) \leq C_1 \cdot \max\{1, \bar{\gamma}(G, \|\cdot\|_X^p)/t\}. \quad (35)$$

We choose $t_0 = \lceil 16C_1^3 \rceil$ let i_0 the smallest i such that $N_i \geq \exp((4t_0)^{1/\delta})$. Denote by $H = B_{i_0}$, $n_0 = N_{i_0}$, and $d_0 = D_{i_0}$. By the choice of the parameters we have $t_0 d_0^{2t_0} \leq n_0 \ll \text{To check the numbers! —Manor}\gg$.

Let $G_0 = \mathcal{C}_{d_0^2}(H)$, and inductively define $G_{i+1} = (\mathcal{C}_{n_0}(\mathcal{A}_{t_0}(G_i))) \mathbb{Z} H$.

First we check that the operations are all feasible. Indeed G_0 has degree d_0^2 . Assume inductively that G_i has degree d_0^2 . Then $\mathcal{A}_{t_0}(G_i)$ has degree $t_0 \cdot d_0^{2(t_0-1)} \leq n_0$. Since both $t_0 \cdot d_0^{2(t_0-1)}$ and n_0 are even, it is feasible to apply $\mathcal{C}_{n_0}(\mathcal{A}_{t_0}(G_i))$ which has degree n_0 . Hence we can apply the zigzag product, since H has n_0 vertices, and indeed G_{i+1} has degree d_0^2 .

Next we prove by induction the bound on $\bar{\gamma}_{G_i} = \bar{\gamma}(G_i, \|\cdot\|_X^p)$. By the assumption on B_i , $\bar{\gamma}_H \leq C_1$. By Lemma 2.12, $\bar{\gamma}_{G_0} \leq C_1$. Assume inductively that $\bar{\gamma}_{G_i} \leq 2C_1^3$. By the choice of t_0 and (35), $\bar{\gamma}_{\mathcal{A}_{t_0}(G_i)} \leq C_1$. By Lemma 2.12, $\bar{\gamma}_{\mathcal{C}_{n_0}(\mathcal{A}_{t_0}(G_i))} \leq 2C_1$, and so by Proposition 4.2, $\bar{\gamma}_{G_{i+1}} \leq 2C_1^3$. $\square$

Theorem 4. *Let X be a p -convex Banach space. Then there exists $C = C(X) > 0$, and a family of $(n_i, 16)$ graphs $\tilde{G}_i = (V_i, E_i)$ where $n_i \nearrow \infty$, such that $\bar{\gamma}(\tilde{G}_i, \|\cdot\|_X^p) \leq C$.*

Proof. Let G_i be the family of degree d_0^2 graphs from Lemma 6.1, and define $\tilde{G}_i = G_i \mathbb{Z} C_n^o$, where C_n^o is the n -vertex cycle with self loops whose degree is 4. By Proposition 2.7, and Proposition 4.2 $\tilde{G}_i$ has degree 16, and $\bar{\gamma}(G_i, \|\cdot\|_X^p) \leq C \cdot 2d_0^{1+2p}$. $\square$

Remark 7. We can reduce the degree to 3, by first adding an arbitrary matching to the graphs of Lemma 6.1, and obtaining a graph with odd degree, and then apply a replacement product on the resulting graph with an odd cycle.

7. MARKOV COTYPE AND LIPSCHITZ EXTENSIONS

For a given stochastic symmetric matrix A , and $\alpha \in (0, 1)$, we denote by

$$\mathcal{G}_\alpha(A) = (1 - \alpha)(I - \alpha A)^{-1} = (1 - \alpha) \sum_{\ell=0}^{\infty} (\alpha A)^\ell.$$

Definition 11 ([5]). A normed space $(X, \|\cdot\|)$ is said to have Markov cotype 2 with constant $K > 0$ if for every $n \times n$ symmetric stochastic matrix A , $\alpha \in (0, 1)$, $x_1, \dots, x_n \in X$

$$\alpha \sum_{i,j} A_{i,j} \|y_i - y_j\|^p + 2(1 - \alpha) \sum_i \|y_i - x_i\|^p \leq K^p (1 - \alpha) \sum_{i,j} \mathcal{G}_\alpha(A)_{i,j} \|x_i - x_j\|^p, \quad (36)$$

where $y_i = \sum_j \mathcal{G}_\alpha(A)_{ij} x_j$.

Ball also gives a metric variant which suffices for his extension theorem.

Definition 12 ([5]). A metric space (X, ρ) is said to have *metric*¹ Markov cotype 2 with constant $K > 0$ if for every $n \times n$ symmetric stochastic matrix A , $\alpha \in (0, 1)$, $x_1, \dots, x_n \in X$ there exist $y_1, \dots, y_n \in X$ such that

$$\alpha \sum_{i,j} A_{i,j} \rho(y_i, y_j)^p + 2(1 - \alpha) \sum_i \rho(y_i, x_i)^p \leq K^p (1 - \alpha) \sum_{i,j} \mathcal{G}_\alpha(A)_{i,j} \rho(x_i, x_j)^p. \quad (37)$$

Note that Markov cotype p implies metric Markov cotype p with the same constant.

Proposition 7.1. *Any p -barycentric space (X, ρ) with constant $K \geq 1$ has metric Markov cotype p with constant $24^{1/p} K$.*

Proof. Fix a asymmetric stochastic $n \times n$ matrix A , $\alpha \in (0, 1)$, and $x_1, \dots, x_n$. From Proposition 5.3, for every $m \in \mathbb{N}$, there exist $y_1^{(m)}, \dots, y_n^{(m)} \in X$,

$$m \sum_{i,j} A_{i,j} \rho(y_i^{(m)}, y_j^{(m)})^p + \sum_i \rho(x_i, y_i^{(m)})^p \leq K^p \sum_{i,j} \mathcal{A}_m(A)_{i,j} \rho(x_i, x_j)^p. \quad (38)$$

¹The notation “metric Markov cotype” is ours. Ball does not bother to give (37) a name.

Define $m_0 = \lfloor 1/(1 - \alpha) \rfloor$ and denote $\bar{e} = \alpha^{-m_0}$. For a given $\ell \in \mathbb{N}$, set $m = (\ell + 1)m_0$. Let $y_i = \text{cm} \left((1 - \bar{e}^{-1})^2 \sum_{\ell=0}^{\infty} (\ell + 1) \bar{e}^{-\ell} y_i^{((\ell+1)m_0)} \right)$. Multiply (38) by $(1 - \bar{e}^{-1})^2 (\ell + 1) \bar{e}^{-\ell}$, and sum up over $\ell \geq 0$. We obtain

$$\begin{aligned}
& m_0 \sum_{ij \in [n]} A_{ij} \rho(y_i, y_j)^p + \sum_{i \in [n]} \rho(x_i, y_i)^p \\
& \leq \sum_{\ell=0}^{\infty} (1 - \bar{e}^{-1})^2 (\ell + 1) \bar{e}^{-\ell} \left((\ell + 1)m_0 \sum_{ij \in [n]} A_{ij} \rho(y_i^{((\ell+1)m_0)}, y_j^{((\ell+1)m_0)})^p + \sum_{i \in [n]} \rho(x_i, y_i^{((\ell+1)m_0)})^p \right) \\
& \leq \sum_{\ell=0}^{\infty} (1 - \bar{e}^{-1})^2 (\ell + 1) \bar{e}^{-\ell} K^p \sum_{i,j \in [n]} \mathcal{A}_{(\ell+1)m_0}(A)_{ij} \rho(x_i, x_j)^p \\
& = K^p \sum_{i,j \in [n]} \sum_{\ell=1}^{\infty} \mathcal{B}_{m_0}(A)_{ij} \rho(x_i, x_j)^p, \quad (39)
\end{aligned}$$

where $\mathcal{B}_{m_0}(A) = \sum_{\ell=0}^{\infty} (1 - \bar{e}^{-1})^2 (\ell + 1) \bar{e}^{-\ell} \mathcal{A}_{(\ell+1)m_0}(A)$.

Next we observe that $\mathcal{B}_{m_0}(A) \sim \mathcal{G}_{\alpha}(A)$. Indeed they are both exponent serieses in A . Fix an exponent $am_0 + b \in \{0, 1, 2, \dots\}$, for $a \in \{0, 1, \dots\}$, and $b \in \{0, 1, \dots, m_0 - 1\}$. The coefficient of A^{am_0+b} in $\mathcal{G}_{\alpha}(A)$ is $(1 - \alpha)\alpha^{am_0+b}$, whereas the coefficient of A^{am_0+b} in $\mathcal{B}_{m_0}(A)$ is

$$(1 - \bar{e}^{-1})^2 \sum_{\ell=a}^{\infty} \frac{(\ell + 1) \bar{e}^{-\ell}}{(\ell + 1)m_0} = (1 - \bar{e}^{-1}) \bar{e}^{-a} / m_0.$$

Compare this two coefficients:

- When $\alpha < 1/2$, then $m_0 = 1$, and $b = 0$, and from the definitions $(1 - \bar{e}^{-1}) \bar{e}^{-a} = (1 - \alpha)\alpha^a$.
- When $1 > \alpha \geq 1/2$, we can approximate $4 \geq \bar{e} \geq \sqrt{e}$, $1/2m_0 \leq 1 - \alpha \leq 1/m_0$, and $\alpha^b \geq \alpha^{m_0} \geq \bar{e}^{-1} \geq 1/4$ So

$$(1 - \alpha)\alpha^{am_0+b} \leq \frac{1}{m_0} \bar{e}^{-a} \leq \frac{1}{1 - 1/\sqrt{e}} (1 - \bar{e}^{-1}) \bar{e}^{-a} / m_0,$$

and on the other hand

$$(1 - \alpha)\alpha^{am_0+b} \geq \frac{1}{2m_0} \bar{e}^{-a} \cdot \frac{1}{4} = \frac{1}{6} (1 - \bar{e}^{-1}) \bar{e}^{-a} / m_0.$$

We have bound term by term the serieses of $\mathcal{G}_{\alpha}(A)$ and $\mathcal{B}_{m_0}(A)$, so we have a bound on the sum of the series so we conclude that for every $i, j \in [n]$

$$\frac{1}{6} \mathcal{B}_{m_0}(A)_{ij} \leq \mathcal{G}_{\alpha}(A)_{ij} \leq \frac{\sqrt{e}}{\sqrt{e} - 1} \mathcal{B}_{m_0}(A)_{ij}$$

Plugging in the last estimate into the right hand side of (39), we conclude that $(X, \|\cdot\|)$ has the metric Markov cotype p property with constant $24^{1/p} K$. $\square$

Ball [5] has proved a Lipschitz extension theorem between space with Markov type 2 and spaces with markov cotype 2. In [5, Section 7] he relaxed the Markov cotype 2 property (Def. 11, which is applicable only in Banach space) into metric Markov cotype 2 (Def. 12). However, in this case he also requires that the metric space satisfies a property he called approximate convexity: A metric space (X, ρ) is approximately convex if there is a $K > 0$ so that for every stochastic $n \times m$ matrix B , symmetric, stochastic $n \times n$ matrix C and $(z_r)_{r=1, \dots, m}$ in X , there are points $(x_i)_{i=1, \dots, n}$ in X with

$$\sum_{ij} C_{ij} \rho(x_i, x_j)^2 + 2 \sum_{i,r} B_{ir} \rho(x_i, z_r)^2 \leq K^2 \sum_{rs} (B^T C B)_{rs} \rho(z_r, z_s)^2.$$

It was observed by Marianna Csörnyei and the second named author that this condition is vacuous. Namely

Proposition 7.2 (Csörnyei–Naor). *Every metric space is approximately convex. $\ll$ What is K ? —Manor $\gg$* MM

This observation is not published and for the sake of completeness we reproduce the proof here with the kind permission of Prof. Csörnyei.

Proof. «Assaf should complete —Manor» □

«TO ADD A PROOF OF THE EXTENSION THEOREM USING THE METRIC MARKOV COTYPE 2 —Manor»

Thus we can conclude

Theorem 5. *Let (X, ρ) be a $CAT(0)$ space, (Y, ρ) a metric space with Markov type 2 with constant K , finite $T, S \subset Y$, and $f : S \rightarrow X$. Then there exists $\hat{f} : T \rightarrow X$ such that $\hat{f}|_{S \cap T} = f|_{S \cap T}$, and $\|\hat{f}\|_{\text{Lip}} \leq O(K)\|f\|_{\text{Lip}}$.*

To better appreciate the extension result in Theorem 5, we list classes of spaces which are known to have the Markov type 2 property:

Theorem 6. *The following classes of metric spaces have Markov type 2:*

- 2-smooth Banach spaces, and in particular L_p for $\infty > p \geq 2$ [20].
- Metric spaces supported on series parallel graphs, and in particular tree metrics [20, 8].

8. LIMITATIONS

In this section we point some obvious points about when things can not be done.

let $p > 0$.

First we point out that $\gamma(G, (\ell_\infty, \|\cdot\|_\infty^p)) \geq (\log_d(n/2))^p$ when G is (n, d) . This is simply because L_∞ contains d_G , and considering the identity mapping, it's clear that for each vertex u at least half of the vertices of G are at distance d_G at least $\log_d(n/2)$ from u .

On the other hand, if the (n, d) graph G is an expander, its diameter is $c \log n$, and by the triangle inequality $\gamma(G, (\ell_\infty, \|\cdot\|_\infty^p)) \leq (c \log n)^p$. So, if G happens also to have large girth, g , then for $t \leq g$ from the same considerations

$$O\left(\frac{\log n}{t}\right)^p \geq \gamma(G, (\ell_\infty, \|\cdot\|_\infty^p)) \geq \Omega\left(\frac{\log n}{t}\right)^p.$$

Few remarks

- For $p < 1$ it shows a space in which the decay of $\gamma(G^t, \cdot)$ is not linear.
- The same work for $\gamma(\mathcal{A}_t(G), \cdot)$
- If one wants to lower than linear decay with power $p > 1$, then all it has to do is to replace $\|\cdot\|_\infty$, with $\|\cdot\|_\infty^{1/3p}$.
- We do not need to invoke the dichotomy for metric transforms from the charlie paper. It's simpler when one only cares about the PI constant.
- Note that $\|\cdot\|_\infty^{1/2}$ has metric cotype 2 (but not with tight m)

I'm not sure this is of interest, one can argue that informally, the decay we show for, say, 100-convex spaces is sub linear. We get linear decay only with power 100. So more interestingly: Can we show a lower bound on the decay of $\gamma(G^t, \|\cdot\|_{100}^2)$? If so, this will explain Ball's comment about ℓ_∞ , since in the limit we won't expect any uniform decay.

BTW, this doesn't preclude the RVW technique, since for it to work we only need a uniform decay.

My feeling is that the answer to my question about the decay of $\gamma(G^t, \|\cdot\|_{100}^2)$ is strongly related to my question about base graph using the ℓ_∞ torus: The candidate graph here is the ℓ_∞ torus. We have the metric cotype inequality that shows the right distortion, which will prove the rate of the decay. The problem is that the metric cotype inequality is not a Poincare inequality (as opposed to the metric type inequality, which can be made Poincare inequality). It shows the significance of my question, whether we can make the metric cotype inequality, a Poincare inequality by taking an appropriate quotient of the ℓ_∞ torus.

9. DISCUSSIONS

When does powers of graphs decay? What are the geometric conditions on ρ for the decay of $\bar{\gamma}(G^t, \rho)$? Is there a “real” difference between the different types of power we have used here? Do we have milder geometric condition on the decay of $\bar{\gamma}(G^t, \rho)$ when G is a Cayley graph of Abelian group? All these questions can be asked about variants of Markov cotype. Do we have decay on all normed spaces?

Under what condition on ρ , for every $M \geq 1$, there exists $t \in \mathbb{N}$, and $s_0 \in [1, \infty)$ such that for every regular graph G , if $\tilde{g}(G, \rho) \geq s_0$, then $\tilde{g}(G^t, \rho) \leq \tilde{g}(G, \rho)/M$. If this is true for every K -convex space, we have an expander construction with respect to K -convex spaces, almost matching Lafforgue's result. Currently, we do not know on any metric space which does not satisfy it to all finite powers.

We know that the class MET^ϵ the decay can not be better than $\bar{\gamma}(G^t, \text{MET}^\epsilon) \geq \bar{\gamma}(G, \text{MET}^\epsilon)/t^\epsilon$. The same goes for all other variants of powers.

Compare variants of Markov cotype.

Base graph based on the tight metric cotype inequality

When does every regular spectral gap translates to bound on $\gamma/\bar{\gamma}$? Average distortion embedding, and present the one for trees.

10. TO DO

The list is not mutually exclusive!

- Can we prove the power variant of the markov cotype?
- Re write Ball's extension theorem using the observations above
- Find base graph for $\text{CAT}(0)$ spaces, and obtain expanders for $\text{CAT}(0)$ spaces.
- Can we have Markov cotype like inequality when the Markov chain is based on the discrete cube, and the host space has (variant of) metric cotype?
- If so, can this be generalized to any abelian graph?
- If so, can it be preserved under some sort of a variant of the zigzag product?
- If this will all work, we may be able to recover Lafforgue totally.
- Understand why our Poincare inequalities for uniformly convex spaces has powers larger than 2. Can we obtain for uniformly convex spaces (2,2) PI? Can this be done using some sort of interpolatuion?
- Find a Cayley expander graphs. The connection between semi-direct product and the zig-zag is not easy to implement. The relevant papers here are [1, 27].
- Show that tightness of the analysis for the zig-zag and tensor product in for the right class of kernels.

APPENDIX A. BALL'S EXTENSION THEOREM

In this section we prove

Theorem 7 (Ball [5]). *Let (Y, σ) be a space with Markov type 2 with constant T and (X, ρ) a space with metric Markov cotype 2 with constant K . Then for any finite $S, T \subset Y$, and any map $f : S \rightarrow X$, there exists a mapping $\hat{f} : T \rightarrow X$, such that $\hat{f}|_{S \cap T} = f|_{S \cap T}$, and $\|\hat{f}\|_{\text{Lip}} \leq O(TK)\|f\|_{\text{Lip}}$.*

MM

«Can we use a compactness argument similar to Rado's compactness theorem (as used by Matousek in the BD-Ramsey paper) that find a "reasonable" completion $\bar{X}$ of X for which we have $\hat{f} : Y \rightarrow \bar{X}$, which is an extension of f and has bounded Lipschitz constant on all finite subsets? —Manor»

Ball proved his extension theorem for X which is Banach space satisfying the stronger condition of Markov cotype 2. At the end of the paper he hints that the Markov cotype 2 condition can be replaced with the metric Markov cotype 2 condition. For the sake of completeness we provide the proof with all the details here.

APPENDIX B. SAND BOX IDEAS

This appendix contains half baked ideas I had, and should probably erased before any one else sees it.

B.1. Actions on $\mathbb{Z}_2^N$. Let $N = 2^n$, and let $\mathbb{Z}_d = \mathbb{Z}/d\mathbb{Z}$. We show how the group $B = \mathbb{Z}_N^{N/n}$ acts on $A = \mathbb{Z}_2^N$ such that the orbit of $e_1 = (1, 0, \dots, 0) \in \mathbb{Z}_2^N$ under the action of B (denoted by $e_1^{\bar{B}}$) gives the set of generators of the power N/n -th power of the Hamming cube.

The action is very simple: For $(\sigma_0, \sigma_1, \dots, \sigma_{N-1}) \in \mathbb{Z}_2^N$ denote by

$$S((\sigma_0, \sigma_1, \dots, \sigma_{N-1}), \ell) = (\sigma_\ell, \sigma_{\ell+1}, \dots, \sigma_{N-1}, \sigma_0, \dots, \sigma_{\ell-1}).$$

Given $\tau = (\tau_0, \tau_1, \dots, \tau_{N/n-1}) \in \mathbb{Z}_N^{N/n}$, and $\sigma \in \mathbb{Z}_2^N$, we write

$$\sigma^{\hat{\tau}} = \sum_{j=0}^{N/n} S(\sigma, \tau_j).$$

THAT'S NONSENSE! $\hat{\tau}$ is not necessarily automorphism.
 $\mathcal{O}$

B.2. Graham-Schmidt Orthogonalization in CAT(0) spaces0. Let (X, ρ) be a CAT(0) space, and $u_1, \dots, u_n \in X$. Let m be the center of mass of $u_1, \dots, u_n$. Define $|u_i| = \rho(m, u_i)$, and $\langle u_i, u_j \rangle = |u_i| \cdot |u_j| \cdot \cos \angle(u_i, m, u_j)$. We next define Graham-Schmidt Orthogonalization (GSO) process as follows: Using "imaginary" orthogonal basis $v_1, \dots, v_n$ for which magnitude $|v_i|^2$ will be well defined (hopefully!) We define it inductively:

$$\begin{aligned} v_1 &= u_1, & |v_1|^2 &= |u_1|^2. \\ v_2 &= u_2 - \frac{\langle u_2, v_1 \rangle}{|v_1|^2} v_1, & |v_2|^2 &= |u_2|^2 - \frac{\langle u_2, v_1 \rangle^2}{|v_1|^2}. \\ v_3 &= u_3 - \frac{\langle u_3, v_2 \rangle}{|v_2|^2} v_2 - \frac{\langle u_3, v_1 \rangle}{|v_1|^2} v_1, & |v_3|^2 &= |u_3|^2 - \frac{\langle u_3, v_2 \rangle^2}{|v_2|^2} - \frac{\langle u_3, v_1 \rangle^2}{|v_1|^2}. \\ &\vdots & &\vdots \\ v_i &= u_i - \sum_{j=1}^{i-1} \frac{\langle u_i, v_j \rangle v_j}{|v_j|^2} & |v_i|^2 &= |u_i|^2 - \sum_{j=1}^{i-1} \frac{\langle u_i, v_j \rangle^2}{|v_j|^2}. \end{aligned}$$

Note that while v_i are not well defined in X , we can define $\langle u_j, v_i \rangle$ by assuming that the inner product is bi-linear, and that the (v_i) are orthogonal. In this case it reduces to a second degree polynomial in $\langle u_k, u_h \rangle$. Note that we also have (synthetically) $|v_i|^2 = \langle v_i, v_i \rangle$

Here is the main conjecture right now.

Conjecture 1. for every i , $|v_i|^2 \geq 0$

With this conjecture we can now define the embedding

$$f_j(u_i) = \frac{\langle u_i, v_j \rangle}{|v_j|} \in \mathbb{R},$$

and $f = (f_1, \dots, f_n) : \{u_1, \dots, u_n\} \rightarrow \mathbb{R}^n$.

From now on, we assume Conj. 1.

Claim 3.

$$u_i = \sum_{j=1}^i \frac{\langle u_i, v_j \rangle v_j}{|v_j|^2},$$

i.e., for any $x \in X$,

$$\langle u_i, x \rangle = \sum_{j=1}^i \frac{\langle u_i, v_j \rangle \langle v_j, x \rangle}{|v_j|^2},$$

Claim 4. For any $i \neq j$, $\langle v_i, v_j \rangle = 0$

Proof. □

Claim 5. If $j > i$ then $\langle u_i, v_j \rangle = 0$

Proof. □

Claim 6.

$$|u_i|^2 = \sum_j \frac{\langle u_i, v_j \rangle^2}{|v_j|^2} = \|f(u_i)\|_2^2.$$

Proof. Again, easy. □

For the above claims we can conclude that

Corollary B.1.

$$\sum_{k=1}^n \frac{\langle u_i, v_k \rangle \cdot \langle u_j, v_k \rangle}{|v_k|^2} = \langle u_i, u_j \rangle.$$

Therefore

$$\|f(u_i) - f(u_j)\|_2^2 = \|f(u_i)\|_2^2 + \|f(u_j)\|_2^2 - 2\langle f(u_i), f(u_j) \rangle \leq |u_i|^2 + |u_j|^2 - 2\langle u_i, u_j \rangle \leq \rho(u_i, u_j)^2$$

At this point we can conclude that f is an average distortion embedding

B.2.1. Example. I want to test the theory developed in this subsection with a simple little example.

The example is the star metric with the center of mass in the center of the star. we can think of it as a metric in ℓ_1^4 :

$$m = (0, 0, 0, 0), \quad u_1 = (1, 0, 0, 0), \quad u_2 = (0, 1, 0, 0), \quad u_3 = (0, 0, 1, 0), \quad u_4 = (0, 0, 0, 1).$$

Then

$$\begin{aligned} v_1 &= u_1 = (1, 0, 0, 0), & |v_1|^2 &= |u_1|^2 = 1. \\ v_2 &= u_2 - \frac{\langle u_2, v_1 \rangle}{|v_1|^2} v_1 = (-1, 1, 0, 0), & |v_2|^2 &= |u_2|^2 - \frac{\langle u_2, v_1 \rangle^2}{|v_1|^2} = 0. \\ v_3 &= u_3 - \frac{\langle u_3, v_2 \rangle}{|v_2|^2} v_2 - \frac{\langle u_3, v_1 \rangle}{|v_1|^2} v_1, & |v_3|^2 &= |u_3|^2 - \frac{\langle u_3, v_2 \rangle^2}{|v_2|^2} - \frac{\langle u_3, v_1 \rangle^2}{|v_1|^2} = 0 \end{aligned}$$

We basically get $v_1 = u_1$, and $v_2 = v_3 = \dots = "0"$. This is a good solution! So this approach is still viable after the star example

SERIOUS PROBLEM: The Graham-Schmidt embedding embed the center of mass to 0, but it does not necessarily maintain the center of mass of the image in 0. In fact it seems that for a star with 3 or more leaves (the example above, it doesn't!

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